

JEE

ADVANCED 2024



Physics

FINAL PRACTICE
PAPER-2



HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PHYSICS

SECTION-I : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. After the switch shown in figure A is closed, there is current i through resistance R . Figure B indicates current variation curves a, b, c and d for four sets of values of R and capacitance C :

(A) R_0 and C_0 , (B) $2R_0$ and C_0 , (C) R_0 and $2C_0$, (D) $2R_0$ and $2C_0$. (B)

Which set goes with which curve? (A)

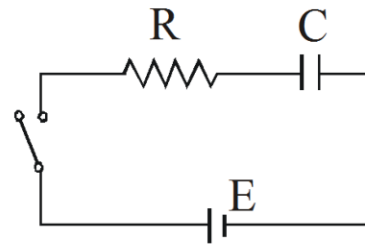


Figure A

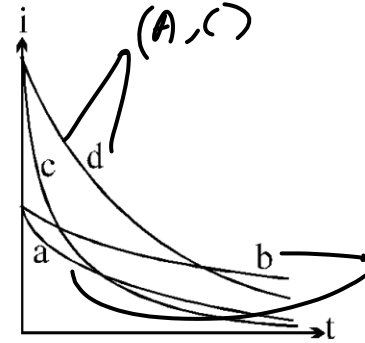


Figure B

$$i = \frac{\varepsilon}{R} e^{-\frac{t}{RC}} \text{ or } i = i_0 e^{-\frac{t}{RC}}$$

$$i_0 = \frac{\varepsilon}{R} \quad \left(\frac{di}{dt} \right) = \frac{\varepsilon}{R} e^{-\frac{t}{RC}} \times \left(-\frac{1}{RC} \right)$$

$$|m| < \left(\frac{1}{R^2 C} \right)$$

(A) a-(A), b-(C), c-(B), d-(D)

(C) a-(D), b-(B), c-(A), d-(C)

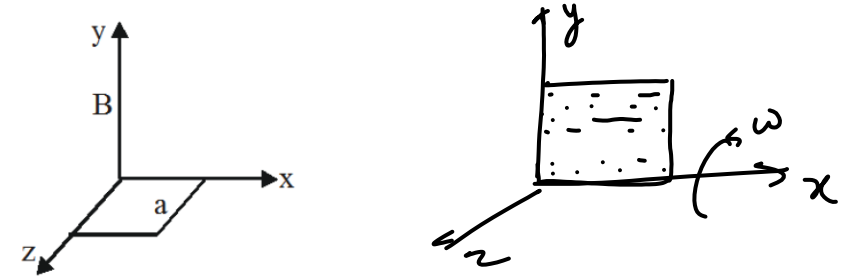
(B) a-(B), b-(D), c-(A), d-(C)

(D) a-(C), b-(A), c-(B), d-(D)

Paragraph for Questions 2

A massless square loop of side a is kept in xz plane as shown. Magnetic field in space is non uniform

given by $\mathbf{B} = \frac{B_0 y}{a} \hat{\mathbf{k}}$. The loop is rotated about x -axis with constant angular velocity ω .



2. Torque required to rotate the loop with constant angular velocity (as a function of time). Take resistance of loop = R .

(A) $\frac{B_0^2 a^4 \omega}{4R} (1 + \sin 2\omega t)^2$

(C) $\frac{4B_0^2 a^4 \omega}{R} \cos^2 \omega t$

(B) $\frac{B_0^2 a^4 \omega}{4R} \sin^2 2\omega t$

(D) $\frac{B_0^2 a^4 \omega}{R} \cos^4 \omega t$

$$Z = \frac{B_0^2 a^4 \sin^2 \omega t}{4 R \omega}$$

$$Z \omega = \frac{\epsilon^2}{R}$$

$$Z = \frac{\epsilon^2}{R \omega}$$

$$d\phi = \frac{B_0 y}{a} dl \cos(\phi_0 - \omega t)$$

$$d\phi = B_0 y dl \sin \omega t$$

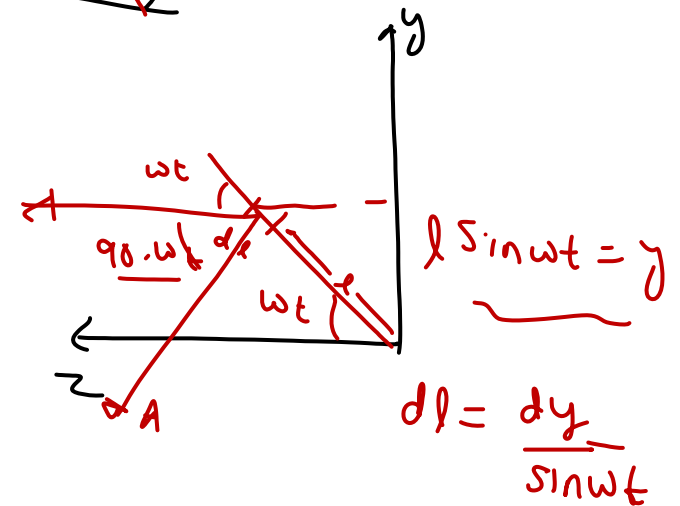
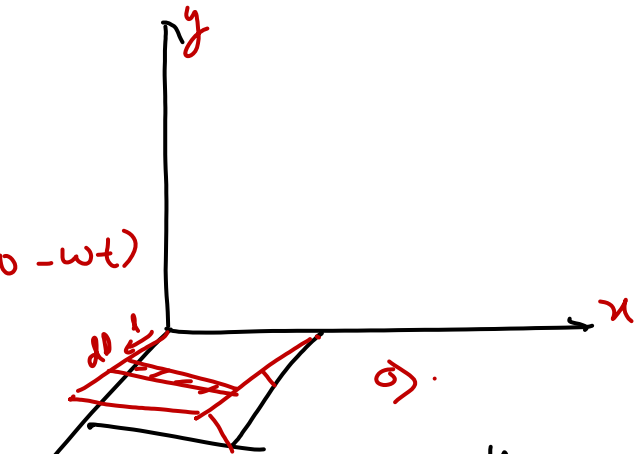
$$\phi = \int B_0 y \frac{dy}{a \sin \omega t} \times \sin \omega t$$

$$= B_0 \int y dy$$

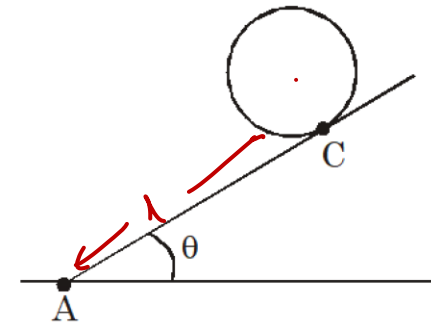
$$= B_0 \left[\frac{y^2}{2} \right]_0^{a \sin \omega t}$$

$$\phi = \frac{B_0}{2} a^2 \sin^2 \omega t$$

$$\epsilon = \frac{d\phi}{dt} = \frac{B_0}{2} a^2 \underbrace{2 \sin \omega t \cos \omega t}_{\sin 2\omega t} \omega = \frac{B_0}{2} a^2 \sin 2\omega t \omega$$



3. A solid sphere of mass 'm' and radius R is released from rest on an inclined plane as shown in figure. The coefficient of friction is $\frac{\tan \theta}{7}$. Find the angular velocity of the sphere at the bottom of the inclined plane (AC = ℓ).



$$\mu_{\min} = \frac{\tan \theta}{1 + \frac{I}{mR^2}}$$

$$= \frac{\tan \theta}{1 + \frac{5}{2}}$$

$$\omega_{\min} = \frac{2 \tan \theta}{7}$$

(A) $\sqrt{\frac{5g\ell \sin \theta}{14R^2}}$

(C) $\sqrt{\frac{25g\ell \sin \theta}{14R^2}}$

~~(B)~~ $\sqrt{\frac{25g\ell \sin \theta}{84R^2}}$

(D) None of these

$$IR = I\alpha$$

$$mgs \sin \theta \cdot R = \frac{2}{5} m R^2 \alpha$$

$$\frac{\sin \theta m g \times R}{7} = \frac{2}{5} m R^2 \alpha$$

$$\frac{5}{14} g \sin \theta = \alpha$$

$$\omega = \alpha t$$

$$\omega = \omega_0 + \alpha t$$

$$a = g \sin \theta - \mu g \cos \theta$$

$$= g \sin \theta - 1 \frac{\sin \theta}{7} g$$

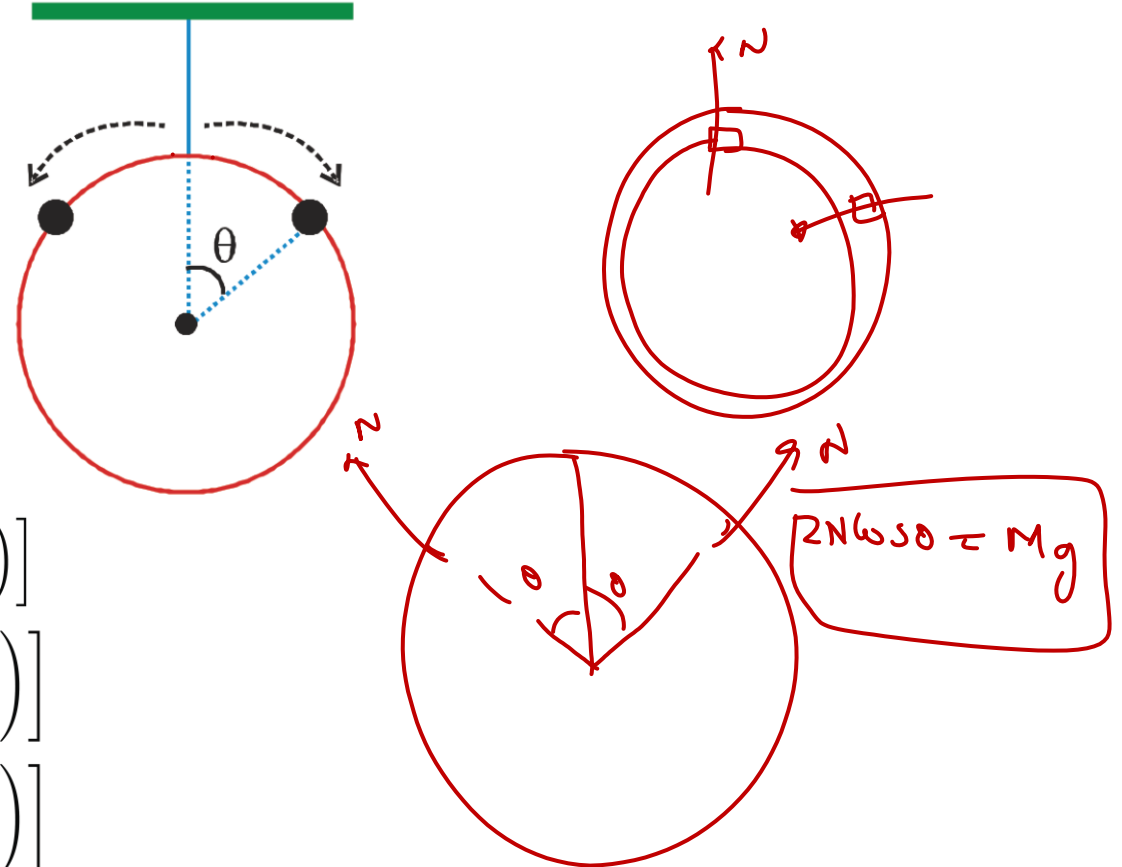
$$a = \frac{6g \sin \theta}{7}$$

A $\rightarrow$ C

$$l = \frac{1}{2} \left(\frac{6g \sin \theta}{7} \right) t^2$$

$$\sqrt{\frac{2l \times 7}{6g \sin \theta}} = t \checkmark$$

4. A smooth circular track of mass M is vertically hung by a string down the ceiling. Two small rings, each of mass m , are initially at rest at the top of the track. They then slide down simultaneously along the track in opposite directions. Find the position of the rings when the tension in the string is zero.



- (A) $\theta = \sin^{-1} \left[\frac{1}{3} \left(1 + \sqrt{1 - \frac{5M}{3m}} \right) \right]$
- (B) $\theta = \cos^{-1} \left[\frac{5}{3} \left(1 + \sqrt{1 - \frac{3M}{2m}} \right) \right]$
- ☒ (C) $\theta = \cos^{-1} \left[\frac{1}{3} \left(1 + \sqrt{1 - \frac{3M}{2m}} \right) \right]$
- (D) $\theta = \cos^{-1} \left[\frac{1}{3} \left(1 + \sqrt{1 - \frac{5M}{3m}} \right) \right]$

$$N + mg \cos \theta = \frac{mv^2}{R}$$

$$2mgR(1 - \cos \theta) = \frac{1}{2}(2m)v^2$$

$$2mg(1 - \cos \theta) = \frac{mv^2}{R}$$

$$N = 2mg(1 - \cos \theta) - mg \cos \theta$$

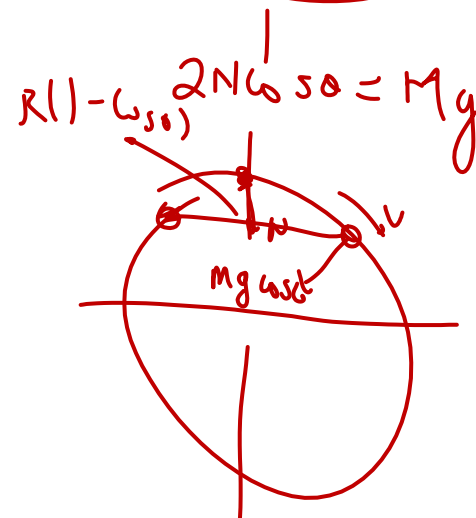
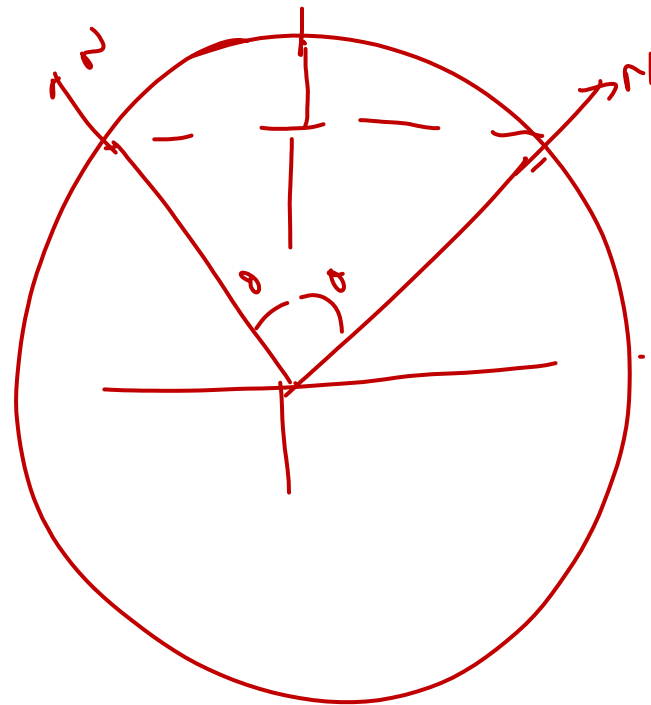
$$N = 2mg - 3mg \cos \theta$$

$$2N \cos \theta = Mg$$

$$4mg \cos \theta - 6mg \cos^2 \theta = Mg$$

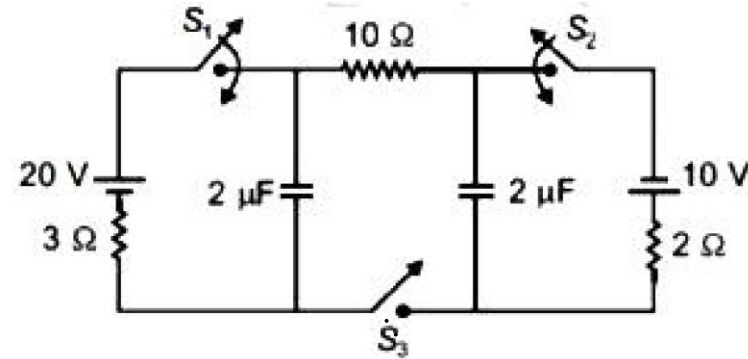
$$6mg \cos^2 \theta - 4mg \cos \theta + Mg = 0$$

$$\cos \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

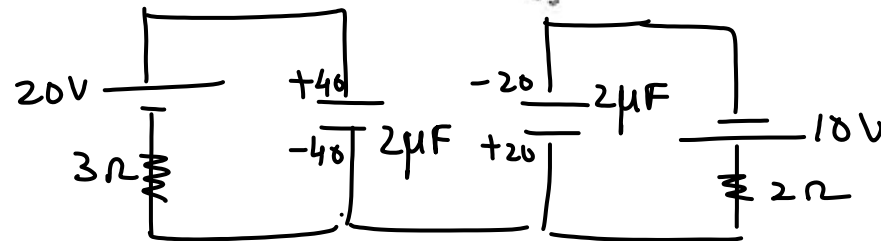


5. In the circuit shown in figure, initially switches S_1 and S_2 were closed and S_3 was open. Now the switches S_1 and S_2 are opened and S_3 is closed. The total heat dissipated by the $10\ \Omega$ resistor is

$$\text{loss} = 450\ \mu\text{J} \\ = \underline{0.45\ \text{mJ}}$$

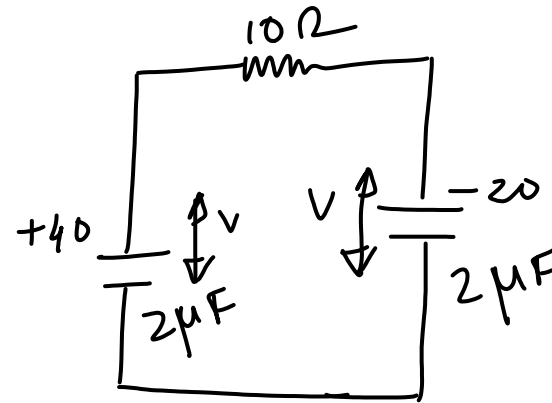


- (A) 0.90 mJ
 ✓ (B) 0.45 mJ
 (C) 1.25 mJ
 (D) 2.50 mJ



$$q = 2 \times 20 \\ = 40\ \mu\text{C}$$

$$U_i = \frac{1}{2} \frac{40^2}{2} + \frac{1}{2} \times \frac{20^2}{2} \\ = 400 + 100 \\ = \underline{500\ \mu\text{J}}$$



$$U_f = \frac{1}{2} \times 2 \times 5^2 \times 2$$

$$= 50 \mu J.$$

$$Q_i = 40 - 20 = 20 \mu C$$

$$Q_f = 2V + 2V = 4V.$$

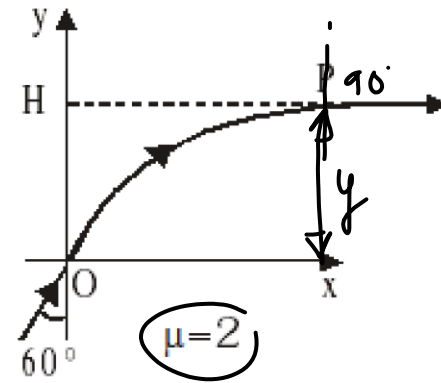
$$Q_i = Q_f \quad 4V = \frac{5}{2} \mu C$$

$$\boxed{V = 5V}$$

$$\Delta U = 500 - 50$$

$$= 450 \mu J$$

6. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \leq y \leq 1$ and $\mu = 2$ for $y < 0$ as shown in figure. A ray of light is incident at origin at an angle 60° with y -axis as shown in the figure. At point P ray becomes parallel to x -axis. The value of H is:-



$$\mu = \frac{2}{1+y^2}$$

(A) $\left\{ \left(\frac{2}{\sqrt{3}} \right) - 1 \right\}^{1/2}$

(B) $\left\{ \frac{2}{\sqrt{3}} \right\}^{1/2}$

(C) $\{ (\sqrt{3}) - 1 \}^{1/2}$

(D) $\left(\frac{4}{\sqrt{3}} - 1 \right)^{1/2}$

$$2 \times \sin 60^\circ = \mu \sin 90^\circ$$

$$2 \times \frac{\sqrt{3}}{2} = \frac{2}{1+y^2} \cdot \sin 90^\circ$$

$$\frac{\sqrt{3}}{2} = \frac{1}{1+y^2}$$

$$\frac{\sqrt{3}}{2} = \frac{1}{1+y^2}$$

$$1+y^2 = \frac{2}{\sqrt{3}}$$

$$y^2 = \frac{2}{\sqrt{3}} - 1$$

$$y = \left(\frac{2}{\sqrt{3}} - 1 \right)^{y_2}.$$

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

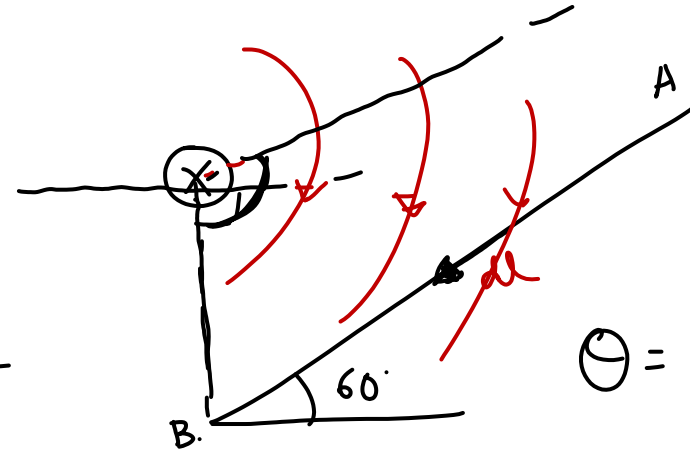
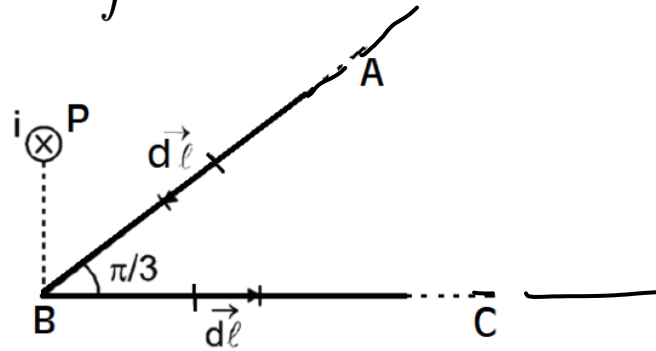
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Line BA & BC are intersecting at 'B' making angle $\pi/3$ at 'B'. Point A & C are at large distance from point B. An infinite wire carrying current 'i' is kept perpendicular to plane containing line BA & BC and intersect the plane at point 'P'. Current in the wire is into the plane of paper as shown in figure.

Then $\int \vec{B} \cdot d\vec{\ell}$ will be equal to:



$$\Theta = \frac{\pi}{2} + \frac{\pi}{3}$$

(A) $-\frac{\mu_0 i}{6}$ on line ABC

☒ (B) $\frac{\mu_0 i}{6}$ i on line ABC

(C) $\frac{2}{2} \mu_0 i$ on line AB

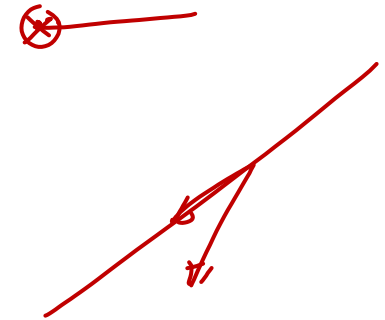
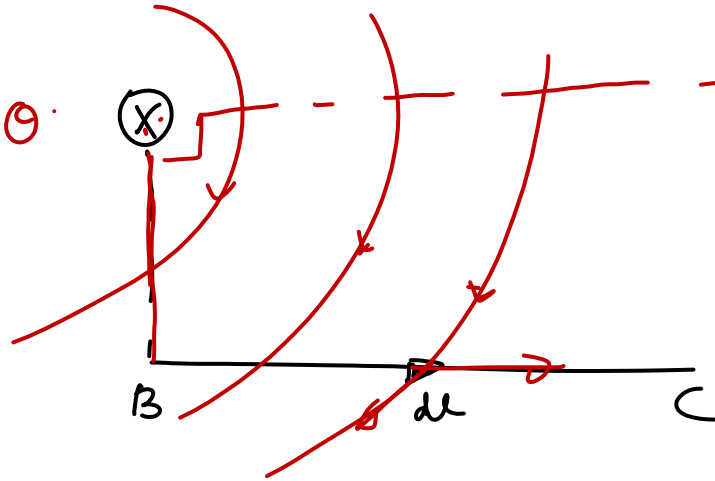
☒ (D) $\frac{5}{12} \mu_0 i$ on line AB

$$\int_{AB} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{2\pi} \cdot \frac{5\pi}{6}$$

$$\theta = \frac{5\pi}{6}$$

$$= \frac{5\mu_0 i}{12}$$

$$\int B \cdot dl = \frac{\mu_0 i}{2\pi} \theta$$



$$\int \vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2\pi} \cdot \frac{\pi}{2}$$

$$= -\frac{\mu_0 i}{4}$$

$$\int_{ABC} \vec{B} \cdot d\vec{l} = \frac{5}{12} \mu_0 i - \frac{\mu_0 i}{4} = \frac{\pi \mu_0 i}{6 + 12} = \frac{\mu_0 i}{6}$$

2. A free hydrogen atom after absorbing a photon of wavelength λ_a gets excited from the state $n = 1$ to the state $n = 4$. Immediately after that the electron jumps to $n = m$ state by emitting a photon of wavelength λ_e . Let the change in momentum of atom due to the absorption and the emission are Δp_a and Δp_e , respectively. If $\lambda_a/\lambda_e = \frac{1}{5}$. Which of the option(s) is/are correct?

[Use $hc = 1242 \text{ eV nm}$; $1 \text{ nm} = 10^{-9} \text{ m}$, h and c are Planck's constant and speed of light, respectively]

~~(A)~~ $\lambda_e = 418 \text{ nm}$

$\frac{\lambda_e}{\lambda_a} = 5$

- ~~(B)~~ The ratio of kinetic energy of the electron in the state $n = m$ to the state $n = 1$ is $\frac{1}{4}$

~~(C)~~ $m = 2$

~~(D)~~ $\Delta p_a/\Delta p_e = \frac{1}{2}$

$\frac{\lambda_a}{\lambda_e} = \frac{1}{5}$

$n=1 \longrightarrow n=4$ λ_a

$p = \frac{h}{\lambda}$

$\downarrow \lambda_e$
 $n=m$

$\frac{p_a}{p_e} = \frac{\lambda_e}{\lambda_a} = 5$

$$\frac{hc}{\lambda_a} = \cancel{13/6} \left[\frac{1}{1^2} - \frac{1}{4^2} \right] \quad 1 \rightarrow 4$$

$$\frac{hc}{\lambda_e} = \cancel{13/6} \left[\frac{1}{m^2} - \frac{1}{4^2} \right] \quad 4 \rightarrow m.$$

$$\frac{\lambda_e}{\lambda_a} = \frac{15/16}{\cancel{1/m^2} - 1/16}$$

$$\frac{1}{m^2} - \frac{1}{16} = \frac{3}{16}$$

$$\frac{1}{m^2} = \frac{\cancel{4}}{16} \frac{1}{4}$$

$$\cancel{5} = \frac{\cancel{15} 3}{16 \left(\frac{1}{m^2} - \frac{1}{16} \right)}$$

$$\boxed{m=2}$$

$$K \propto \frac{Z^2}{n^2}$$

$$\begin{matrix} n_1 = 2 \\ n_2 = 1 \end{matrix} \cdot \frac{K_1}{K_2} = \left(\frac{n_2}{n_1} \right)^2 = \left(\frac{1}{2} \right)^2 = \frac{1}{4}$$

$$\cdot \frac{hc}{\lambda_e} = 13.6 \left[\frac{1}{2^2} - \frac{1}{4^2} \right]$$

$$= 13.6 \left[\frac{1}{4} - \frac{1}{16} \right]$$

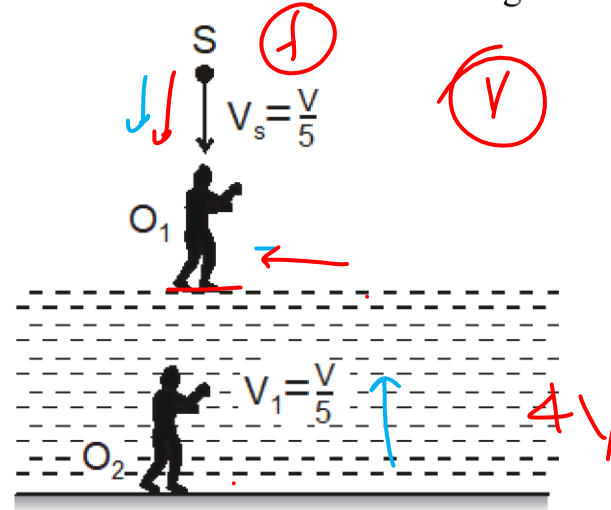
$$\frac{hc}{\lambda_e} = 13.6 \times \frac{3}{16}$$

$$\lambda_e = \frac{hc \times 16}{3 \times 13.6}$$

$$= \frac{12400 \times 16}{3 \times 13.6}$$

$$= \underline{4862 \text{ \AA}}$$

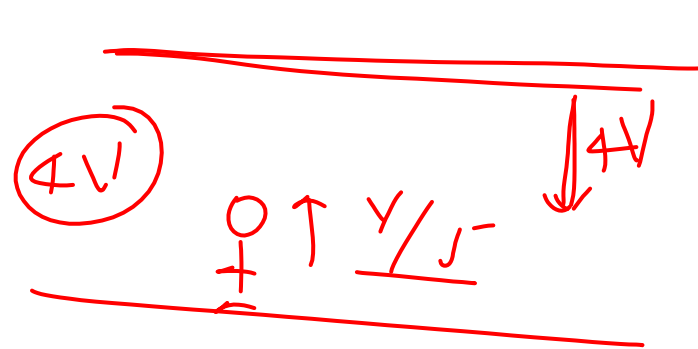
3. In the figure shown an observer O_1 floats (static) on water surface with ears in air while another observer O_2 is moving upwards with constant velocity $V_1 = V/5$ in water. The source moves down with constant velocity $V_s = V/5$ and emits sound of frequency 'f'. The velocity of sound in air is V and that in water is $4V$. For the situation shown in figure :



- (A) ~~The wavelength of the sound received by O_1 is $\frac{4V}{5f}$~~
- (B) ~~The wavelength of the sound received by O_1 is V/f~~
- (C) ~~The frequency of the sound received by O_2 is $\frac{21f}{16}$~~
- (D) ~~The wavelength of the sound received by O_2 is $\frac{16V}{5f}$~~

$$\begin{aligned}
 f_o &= \left(\frac{V \pm V_o}{V \pm V_s} \right) f \\
 &= \frac{V}{(V - V_s)} f \\
 &= \frac{V}{(V - V/5)} f \\
 &= \frac{5V}{4V} f \\
 \lambda_s &= \frac{4V}{5f}
 \end{aligned}$$

$$\rightarrow 0 \downarrow V/5$$



$$f_a = f_w$$

$$\frac{V}{\lambda_a} = \frac{4V}{\lambda_w}$$

$$\lambda_w = \frac{4 \lambda_a}{5} = 4 \times \frac{4V}{5f}$$

$$= \frac{16V}{5f}$$

$$f_{o2} = \frac{4V + V/5}{\lambda_w}$$

$$= \frac{21V}{5 \times \frac{16V}{5f}} = \frac{21f}{16}$$

4. The filament of a light bulb has surface area 64 mm^2 . The filament can be considered as a black body at temperature 2500 K emitting radiation like a point source when viewed from far. At night the light bulb is observed from a distance of 100 m . Assume the pupil of the eyes of the observer to be circular with radius 3 mm . Then

(Take Stefan-Boltzmann constant $= 5.67 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4}$, Wien's displacement constant $= 2.90 \times 10^{-3} \text{ m-K}$, Planck's constant $= 6.63 \times 10^{-34} \text{ Js}$, speed of light in vacuum $= 3.00 \times 10^8 \text{ ms}^{-1}$)-

- ~~(A)~~ power radiated by the filament is in the range 642 W to 645 W
- ~~(B)~~ radiated power entering into one eye of the observer is in the range $3.15 \times 10^{-8} \text{ W}$ to $3.25 \times 10^{-8} \text{ W}$
- ~~(C)~~ the wavelength corresponding to the maximum intensity of light is 1160 nm
- ~~(D)~~ taking the average wavelength of emitted radiation to be 1740 nm , the total number of photons entering per second into one eye of the observer is in the range 2.75×10^{11} to 2.85×10^{11}

$$P_r = \sigma e A T^4 = \sigma A T^4$$

$$= 5.67 \times 10^{-8} \times 64 \times 10^{-6} \times (2500)^4$$

$$= 141$$



$$P_r = I \cdot A$$

$$= \frac{P_r}{(4\pi r^2)} (\pi \times 3 \times 10^{-3})^2$$

$$= \frac{141}{4\pi \times (80)^2} \times (\pi \times 3 \times 10^{-3})^2$$

$$\lambda_m T = b = 3.2 \times 10^{-8}$$

$$\lambda_m = \frac{2.90 \times 10^{-3}}{2500} \approx 1160 \text{ nm}$$

$$\left(\frac{N}{s}\right) (E_{\text{photon}}) = \left(\frac{E_T}{s}\right) = P_{\text{received}}$$

$$\left(\frac{N}{s}\right) = \frac{3.2 \times 10^{-8} \text{ (A)}}{hc}$$

$$= \frac{3.2 \times 10^{-8} \times 1740 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^8}$$

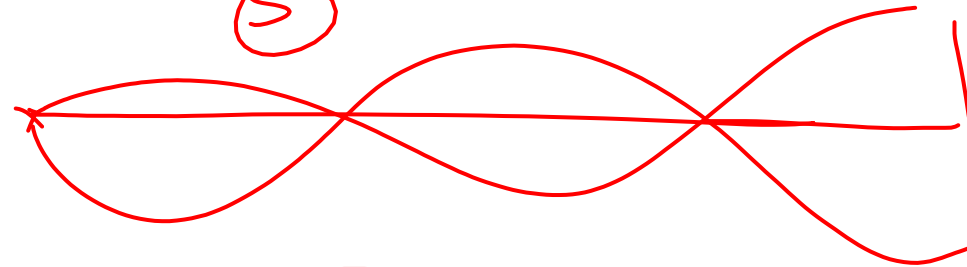
5. One end of a taut string of length 3m along the x-axis is fixed at $x = 0$. The speed of the waves in the string is 100 ms^{-1} . The other end of the string is vibrating in the y direction so that stationary waves are set up in the string. The possible waveform(s) of these stationary waves is (are)

(A) $y(t) = A \sin \frac{\pi x}{6} \cos \frac{50\pi t}{3}$ ✓

(B) $y(t) = A \sin \frac{\pi x}{3} \cos \frac{100\pi t}{3}$ ✗

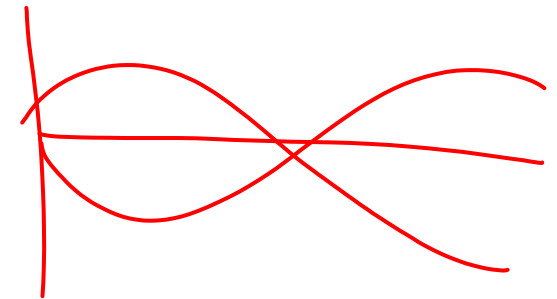
(C) $y(t) = A \sin \frac{5\pi x}{6} \cos \frac{250\pi t}{3}$ ✓

(D) $y(t) = A \sin \frac{5\pi x}{2} \cos 250\pi t$ ✓



Handwritten calculations:
 $\frac{5\pi}{2} = \frac{\pi}{\lambda} \cdot 3$
 $\lambda = \frac{6}{5}$
 $n = 15$

Handwritten text: A, C, D



$$y = A \sin(k)x \sin \omega t$$

$$k = \frac{2\pi}{\lambda}$$



$$= \frac{2\pi(2n+1)}{\lambda}$$

$$k = \frac{1}{\lambda} \frac{2\pi}{\lambda} (2n+1)$$

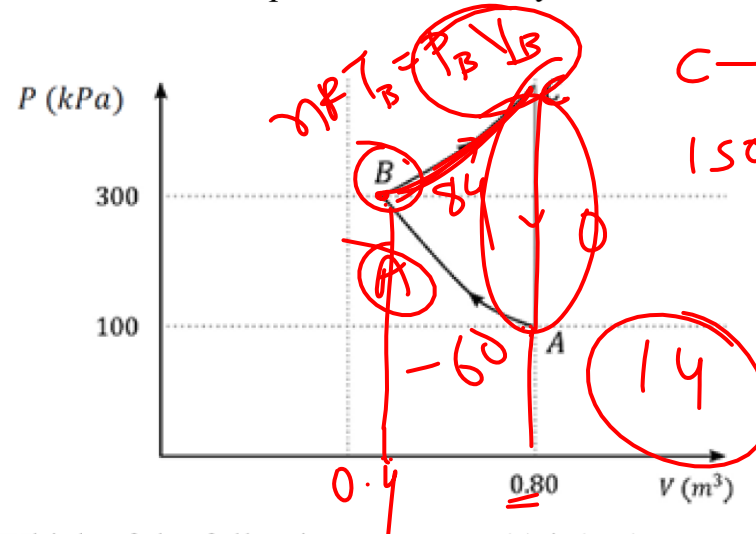
$$(2n+1) \frac{\lambda}{4} = l$$

$$\lambda = \frac{4l}{(2n+1)} = \frac{12}{(2n+1)}$$

$$\begin{aligned}
 \omega &= 2\pi f \\
 &= 2\pi \frac{v}{\lambda} \\
 &= \cancel{2\pi} \times \frac{50}{\cancel{126}} (2n+1) \\
 &= \frac{50\pi}{3} (2n+1)
 \end{aligned}$$

6.

In the given P-V diagram, a monoatomic gas ($\gamma = \frac{5}{3}$) is first compressed adiabatically from state A to state B. Then it expands isothermally from state B to state C. [Given: $\left(\frac{1}{3}\right)^{0.6} \approx 0.5$, $\ln 2 \approx 0.7$].



Handwritten calculations:

$$W = \left(\frac{f}{2}\right) (P_A V_A - P_B V_B)$$

$$= \frac{3}{2} (100 \times 0.8 - 300 \times 0.4)$$

$$= -60$$

$$P_A V_A^\gamma = P_B V_B^\gamma$$

Which of the following statement(s) is(are) correct?

- ☒ (A) The magnitude of the total work done in the process $A \rightarrow B \rightarrow C$ is 144 kJ.
- ☒ (B) The magnitude of the work done in the process $B \rightarrow C$ is 84 kJ.
- ☒ (C) The magnitude of the work done in the process $A \rightarrow B$ is 60 kJ.
- ☒ (D) The magnitude of the work done in the process $C \rightarrow A$ is zero.

Handwritten calculation for (B):

$$\left(\frac{V_B}{V_A}\right)^\gamma = \frac{P_A}{P_B}$$

$$= \frac{1}{3}$$

$$\left(\frac{V_B}{V_A} \right)^4 = \frac{1}{3}$$

$$\frac{V_B}{V_A} = \left(\frac{1}{3} \right)^{1/4}$$

$$V_B = V_A \left(\frac{1}{3} \right)^{3/5}$$

$$= 0.8 \times \left(\frac{1}{3} \right)^{0.6} = 0.5 \times 0.8 = 0.4$$

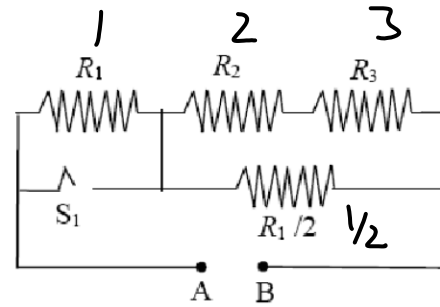
B $\rightarrow$ C (isothermal)

$$W = nRT \ln \left(\frac{V_C}{V_B} \right)$$

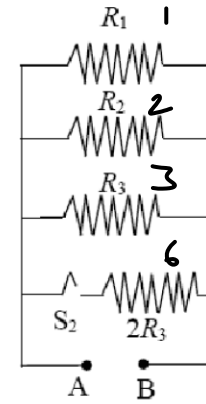
$$= 0.4 \times 300 \ln 2$$

$$= \frac{120 \times 0.7}{84} \text{ kJ}$$

7. In Circuit-1 and Circuit-2 shown in the figures, $R_1 = 1 \Omega$, $R_2 = 2 \Omega$ and $R_3 = 3 \Omega$. P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively. Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively.



Circuit-1



Circuit-2

Which of the following statement(s) is(are) correct?

- (A) When a voltage source of 6 V is connected across A and B in both circuits, $P_1 < P_2$.
- (B) When a constant current source of 2 Amp is connected across A and B in both circuits, $P_1 > P_2$.
- (C) When a voltage source of 6 V is connected across A and B in Circuit-1, $Q_1 > P_1$.
- (D) When a constant current source of 2 Amp is connected across A and B in both circuits, $Q_2 < Q_1$.

when switch is open

$$R_{eq1} = \frac{5 \cdot \frac{1}{2}}{5 + \frac{1}{2}} + 1 = \frac{5}{11} + 1 = \frac{16}{11} \Omega$$

$$R_{eq2} = \frac{6}{6 + 2 + 3} = \frac{6}{11} \Omega$$

when switch is closed

$$R'_{eq1} = \frac{5}{11} \Omega$$

$$R'_{eq2} = \frac{\frac{6}{11} \times 6}{\frac{6}{11} + 6} = \frac{36}{72/11} = \frac{1}{2} \Omega$$

When Voltage Source = 6V

$$P_1 = \frac{V^2}{R_{eq1}} = \frac{6^2}{16} \times 11 = \frac{99}{4} W$$

$$P_2 = \frac{V^2}{R_{eq2}} = \frac{6^2}{6/11} = 6 \times 11$$

$$Q_1 > P_1$$

$$Q_1 = \frac{V^2}{R'_{eq1}} = \frac{6^2}{5/11} = \frac{36 \times 11}{5} W$$

$$Q_2 = \frac{V^2}{R'_{eq2}} = \frac{6^2}{1/2} = 72 W$$

When Current Source = 2A

$$Q_2 > Q_1$$

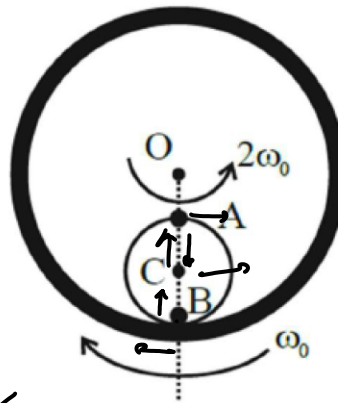
$$P_1 = i^2 \cdot R_{eq1} = 4 \times \frac{16}{11} = \frac{64}{11} W$$

$$P_2 = i^2 R_{eq2} = 4 \times \frac{6}{11} = \frac{24}{11} W$$

$$Q_1 = i^2 R'_{eq1} = 4 \times \frac{5}{11} = \frac{20}{11} W$$

$$Q_2 = i^2 R'_{eq2} = 4 \times \frac{1}{2} = 2 W$$

8. A hollow cylinder of radius $4R$ is rotating about fixed horizontal axis passing through point 'O' with angular velocity ω_0 . A solid cylinder of radius ' R ' is rolling without slipping with respect to inner surface of hollow cylinder. At the given instant, the line OC has angular velocity of $2\omega_0$. Point A and B are topmost and bottom most point of solid cylinder respectively and C is its centre. Then, at the given instant.



$$\omega R - v_c = v$$

$$\omega R - 2\omega_0(3R) = \omega_0(4R)$$

$$\Rightarrow \omega = 10\omega_0$$

- (A) Angular velocity of solid cylinder about its own axis is $10\omega_0$
- (B) Magnitude of acceleration of point A of solid cylinder is $88\omega_0^2 R$
- (C) Magnitude of acceleration of point B of solid cylinder with respect to contact surface of hollow cylinder is $12\omega_0^2 R$
- (D) Radius of curvature of point A is $\frac{32R}{11}$

$$\begin{aligned}
 \vec{a}_A &= \vec{a}_{Ac} + \vec{a}_c \\
 &= (10\omega_0)^2 R \downarrow + (2\omega_0)^2 3R \uparrow \\
 &= (100 - 12) \omega_0^2 R \downarrow \\
 &= 88 \omega_0^2 R \downarrow
 \end{aligned}$$

$$\begin{aligned}
 \vec{v}_A &= \vec{v}_{Ac} + \vec{v}_c \\
 &= 10\omega_0 R \rightarrow + 6\omega_0 R \rightarrow \\
 &= 16\omega_0 R \rightarrow
 \end{aligned}$$

$$R_c = \frac{V_A^2}{a_A} = \frac{(16\omega_0 R)^2}{88\omega_0^2 R} = \frac{32}{11} R$$

$$\begin{aligned}
 \vec{G}_B &= \vec{G}_{BC} + \vec{G}_C \\
 &= (10\omega_0)^2 R \uparrow + 12\omega_0^2 R \uparrow \\
 &= 112\omega_0^2 R \uparrow
 \end{aligned}$$

$$\begin{aligned}
 \vec{G}_{BS} &= \vec{G}_B - \vec{G}_S \\
 &= 112\omega_0^2 R \uparrow - 4\omega_0^2 R \uparrow \\
 &= 108\omega_0^2 R \uparrow
 \end{aligned}$$

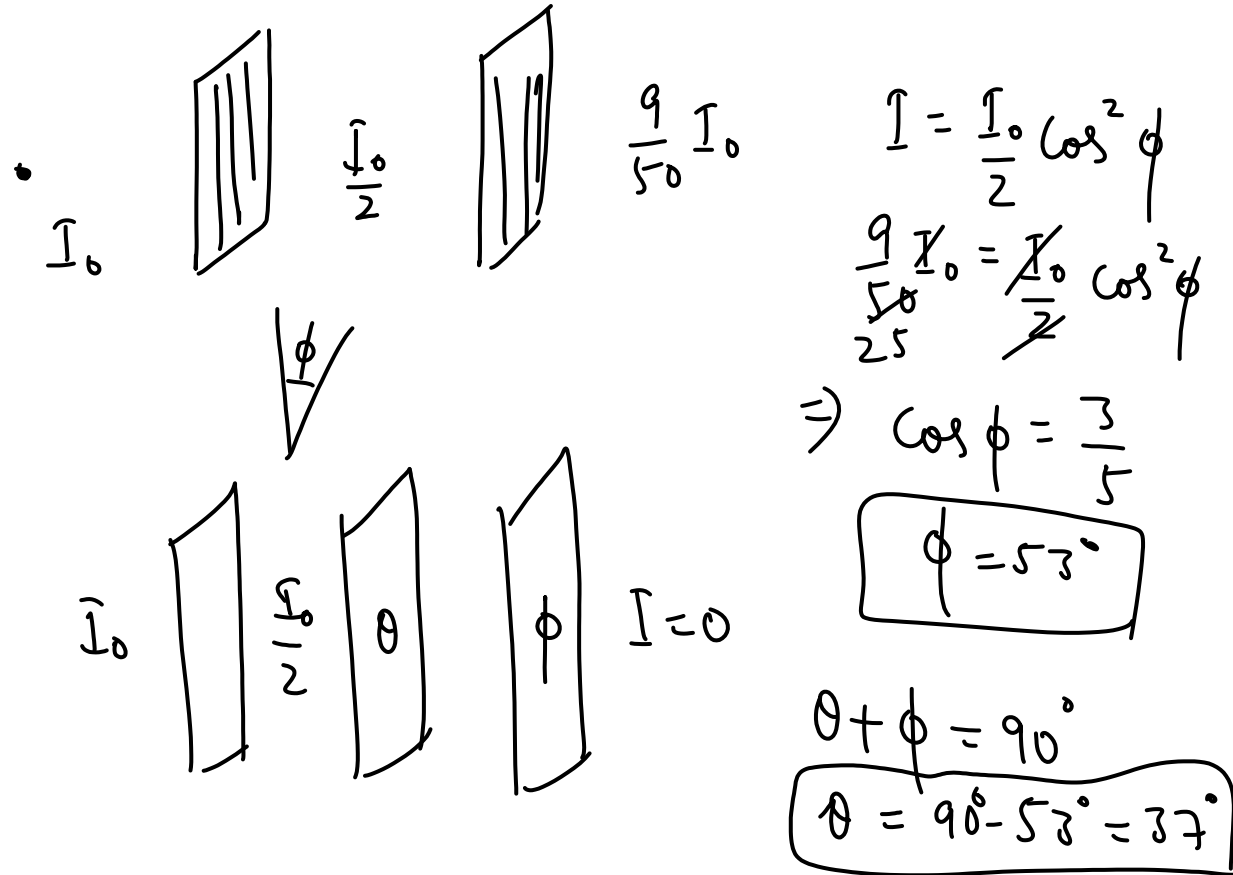
SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33 , $-.30$, 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that the intensity of light emitted from the second polaroid is $\frac{9}{50} I_0$. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids, such that the intensity of light emerging from the last polaroid is zero, then angle θ (in degree) is :



The diagram illustrates two experimental setups for polarized light. In the top setup, an unpolarized beam of intensity I_0 passes through a vertical polaroid, resulting in vertically polarized light of intensity $\frac{I_0}{2}$. This light then passes through a second vertical polaroid, resulting in an intensity of $\frac{9}{50} I_0$. In the bottom setup, the same unpolarized beam of intensity I_0 passes through a vertical polaroid to become vertically polarized with intensity $\frac{I_0}{2}$. This light then passes through a third polaroid whose axis is at an angle θ to the vertical. Finally, it passes through a fourth polaroid whose axis is at an angle ϕ to the vertical, resulting in zero intensity ($I=0$).

Handwritten calculations:

$$I = \frac{I_0}{2} \cos^2 \phi$$

$$\frac{9}{50} I_0 = \frac{I_0}{2} \cos^2 \phi$$

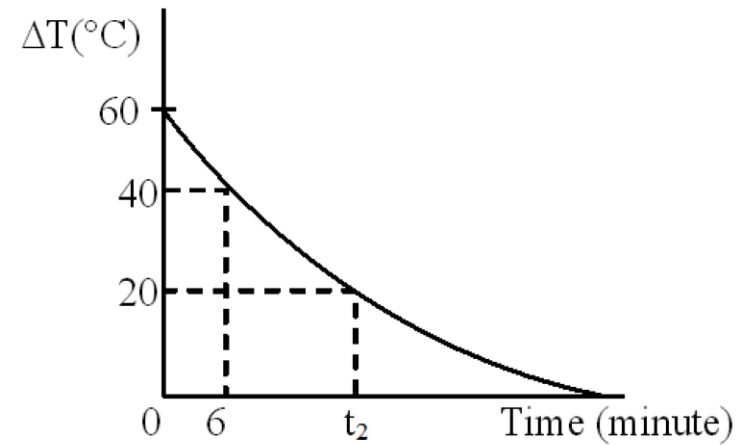
$$\Rightarrow \cos \phi = \frac{3}{5}$$

$$\phi = 53^\circ$$

$$\theta + \phi = 90^\circ$$

$$\theta = 90^\circ - 53^\circ = 37^\circ$$

2. In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C . The value of t_2 as mentioned in the graph will be _____.



$$\Delta T = T - T_s = (T_i - T_s) e^{-kt}$$

At $t = 0$

$$60 = (80 - T_s)$$

$$\Rightarrow T_s = 20^\circ\text{C}$$

$$40 = 60 e^{-k \cdot 6}$$

$$\Rightarrow \frac{2}{3} = e^{-k \cdot 6} \quad \text{--- (1)} \Rightarrow \ln \frac{2}{3} = -k \cdot 6$$

$$20 = 60 e^{-k t_2}$$

$$\Rightarrow \frac{1}{3} = e^{-k t_2} \Rightarrow \ln \frac{1}{3} = -k t_2$$

$$t_2 = \frac{6}{0.37} = 16.21 \dots$$

$$\frac{\ln \frac{2}{3}}{\ln \frac{1}{3}} = \frac{6}{t_2}$$

$$\Rightarrow \frac{\ln 3/2}{\ln 3} = \frac{6}{t_2} \Rightarrow \frac{\ln 3 - \ln 2}{\ln 3} = \frac{6}{t_2}$$

$$\Rightarrow 1 - \frac{\ln 2}{\ln 3} = \frac{6}{t_2}$$

$$\Rightarrow 1 - \frac{0.693}{1.1} = 1 - 0.63 = 0.37 = \frac{6}{t_2}$$

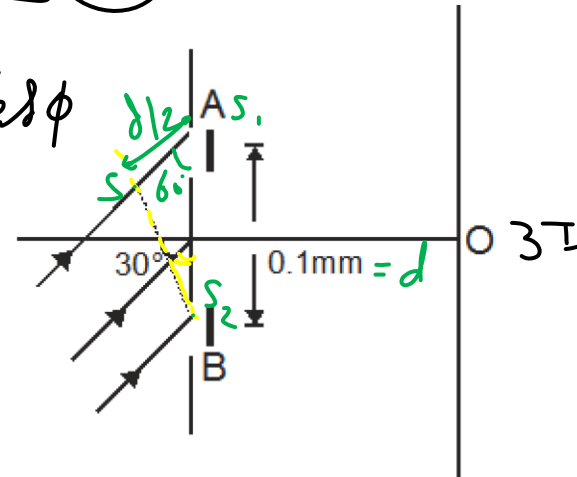
3. In a YDSE a parallel beam of light of wavelength $6000\text{\AA}$ is incident on slits at angle of incidence 30° . A and B are two thin transparent films each of refractive index 1.5. Thickness of A is $20.4\text{ }\mu\text{m}$. Light coming through A and B have intensities I and $4I$ respectively on the screen. Intensity at point O which is symmetric relative to the slits is $3I$. The central maxima is above O. What is the maximum thickness of B (in μm).

$$3I = I + 4I + 2\sqrt{I}\sqrt{4I}\cos\phi$$

$$\Rightarrow -\frac{1}{2} = \cos\phi$$

$$\Rightarrow \cos\phi = -\frac{1}{2}$$

$$\Rightarrow \boxed{\phi = \frac{2\pi}{3}}$$



$$\phi = \frac{2\pi}{\lambda} (\mu_2 t_2 - \mu_1 t_1)$$

$$\frac{2\pi}{3} = \frac{2\pi}{\lambda} \left[\frac{d}{2} + t_n(n-1) - t_B(n-1) \right] \quad \lambda(n-1)$$

$$\Rightarrow \frac{1}{3} = \frac{d}{2} + \frac{t_n}{2} - \frac{t_B}{2}$$

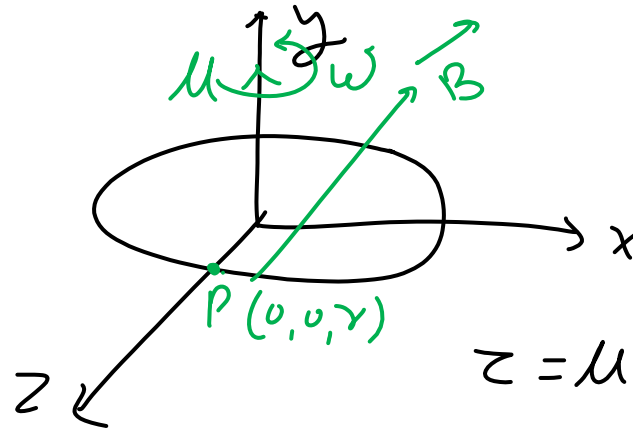
$$\Rightarrow t_B = d + t_n - \frac{2}{3}$$

$$= 0.1 \times 10^3 + 20.4 - \frac{2 \times \overset{2000}{\cancel{6000}} \times 10^{-4}}{3}$$

$$= 100 + 20.4 - 0.4$$

$$\boxed{t_B = 120 \mu\text{m}}$$

4. A disk of radius r , having homogenous mass and charge distribution m and Q respectively, is rotating on its own axis with $(\omega)\hat{j}$ in zx plane with centre at origin in a free space (no hinge). P is a point on the circumference of disk. At an instant when P is at $(0, 0, r)$, a magnetic field $\left[\sqrt{n} \frac{\omega m}{Q} \right] (-\hat{k})$ is switched on in the region. As a result, the acceleration of P becomes double momentarily. Write n .



$$\mu = \frac{q}{2m} (L) = \frac{q}{2m} \left(\frac{m r^2 \omega}{2} \right) = \frac{\omega r^2 q}{4}$$

$$\tau = \mu B = I \alpha$$

$$\Rightarrow \alpha = \frac{\mu B}{I} = \frac{\omega r^2 q B}{4 \left(\frac{m r^2}{2} \right)} = \frac{q B \omega}{m}$$

$$a_z = r \alpha = \frac{q B \omega r}{m}$$

$$2a_c = \sqrt{a_t^2 + a_c^2}$$

$$\Rightarrow a_t = \sqrt{3} a_c$$

$$\Rightarrow \frac{q B \cancel{\omega}}{m} = \sqrt{3} \cancel{\omega}$$

$$\Rightarrow \boxed{B = \frac{\sqrt{3} \omega m}{q}}$$

$$\boxed{n=3}$$

5. A sinusoidal progressive wave is generated in a string. It's equation is given by

$$y = (2\text{mm}) \sin(2\pi x - 100\pi t + \pi/3)$$

The time when particle at $x = 4\text{ m}$ first passes through mean position, will be $\frac{1}{n \times 10}$ s. Find n

$$\boxed{n = 30}$$

$$0 = 2 \sin\left(8\pi - 100\pi t + \frac{\pi}{3}\right)$$

$$\Rightarrow 0 = \sin\left(-100\pi t + \frac{\pi}{3}\right)$$

$$\Rightarrow n\pi = -100\pi t + \frac{\pi}{3}$$

$$\Rightarrow 100t = \frac{1}{3}$$

$$\Rightarrow t = \frac{1}{300} = \frac{1}{n \times 10}$$

30

6. In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is 41.

$$\begin{aligned}
 49 \text{ MSD} &= 50 \text{ VSD} \Rightarrow LC = 1 \text{ MSD} - 1 \text{ VSD} \\
 &= 1 \text{ MSD} - \frac{49}{50} \text{ MSD} \\
 &= \left(1 - \frac{49}{50}\right) \text{ MSD} \\
 &= \frac{1}{50} \times \frac{1}{20} \text{ cm} \\
 &= \frac{1}{1000} \text{ cm} = 0.01 \text{ mm}
 \end{aligned}$$

$$d' = \frac{d}{\mu} \Rightarrow \boxed{\mu = \frac{d}{d'}}$$

$$\Rightarrow \frac{\Delta \mu}{\mu} = \frac{\Delta d}{d} + \frac{\Delta d'}{d'}$$

$$\begin{aligned}
 \Delta \mu &= \left(\frac{\Delta d}{d} + \frac{\Delta d'}{d'} \right) \mu \\
 &= \left(\frac{0.01}{5.25} + \frac{0.01}{5} \right) \frac{5.25}{5} \\
 &= \left(\frac{1}{525} + \frac{1}{500} \right) \frac{525}{500} \\
 &= \left(1 + \frac{525}{500} \right) \frac{1}{500} \\
 &= \frac{41}{20} \left(\frac{1}{500} \right)^{20} = \frac{41}{10 \times 10^3} \\
 &= \frac{41}{10} \times 10^{-3}
 \end{aligned}$$

JEE

ADVANCED 2024



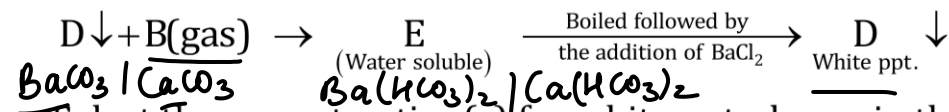
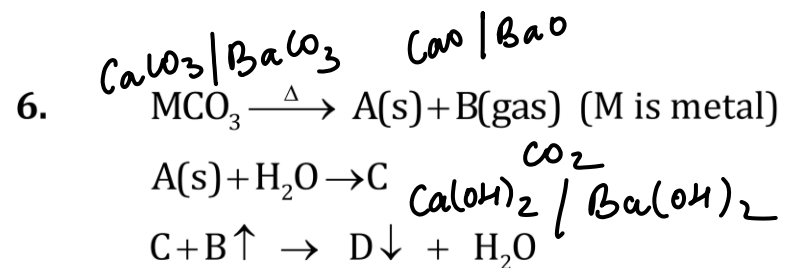
FINAL PRACTICE
PAPER-2



SECTION-II : (Maximum Marks : 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +4	If only (all) the correct option(s) is(are) chosen;
<i>Partial Marks</i>	: +3	If all the four options are correct but ONLY three options are chosen;
<i>Partial Marks</i>	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct;
<i>Partial Marks</i>	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
<i>Zero Marks</i>	: 0	If unanswered;
<i>Negative Marks</i>	: -2	In all other cases.
- For example, if first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.



Select the correct option(s) for white ppt. shown in the above reactions :

☒ (A) CaCO₃

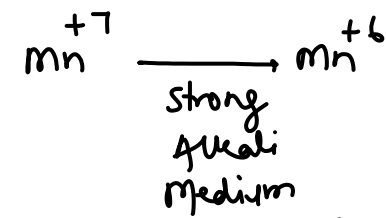
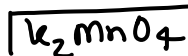
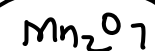
(B) K₂CO₃ ☒

☒ (C) BaCO₃

(D) Na₂CO₃ ☒

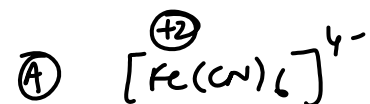
BaCO_3
 CaCO_3 } white ppt

7. K_2MnO_4 is formed when
- ☒ (A) chlorine is passed in an aq. KMnO_4 solution
 - ☒ (B) manganese dioxide is fused with KOH in air
 - ☒ (C) formaldehyde reacts with potassium permanganate in presence of a strong alkali.
 - ☒ (D) potassium permanganate reacts with conc. sulphuric acid



8. Which of the following statements is/are true?

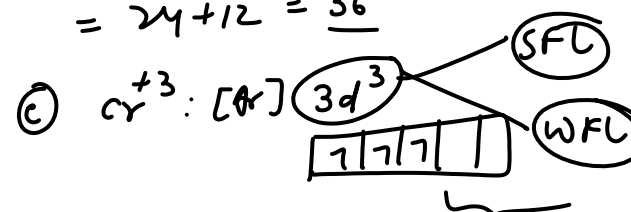
- ~~(A)~~ In Ferrocyanide ion, the effective atomic number is 36
~~(B)~~ Chelating ligands are atleast bidentate ligands
~~(C)~~ ⁽⁺²⁾ $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^\ominus$ and ⁽⁺³⁾ $[\text{CrCl}_3(\text{NH}_3)_3]$ both have d^2sp^3 hybridisation
~~(D)~~ As the number of rings in complex increases, stability of complex (chelate) also increases



$$\text{EAN} = 26 - 2 + 6 \times 2 = 24 + 12 = 36$$

$(\text{CN})^-$
 ↪ monodentate ligand
 ↪ $2e^-$ donor

③ chelation
 ↪ Ring formation



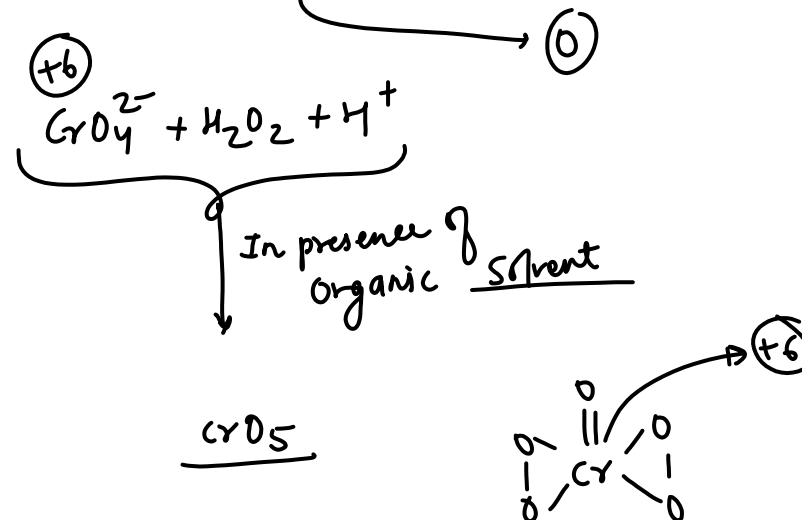
SECTION-III: (Maximum Marks: 18)

- This section contains **SIX (06)** question. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +3	If ONLY the correct numerical value is entered.
<i>Zero Marks</i>	: 0	In all other cases.

4. Chromate solution + $\text{H}_2\text{O}_2 + \text{H}^+$ $\xrightarrow[\text{organic solvent}]{\text{In presence of}}$ permanent deep blue coloured compound.

Find the change in oxidation state of Cr-atom in the reactant and product.



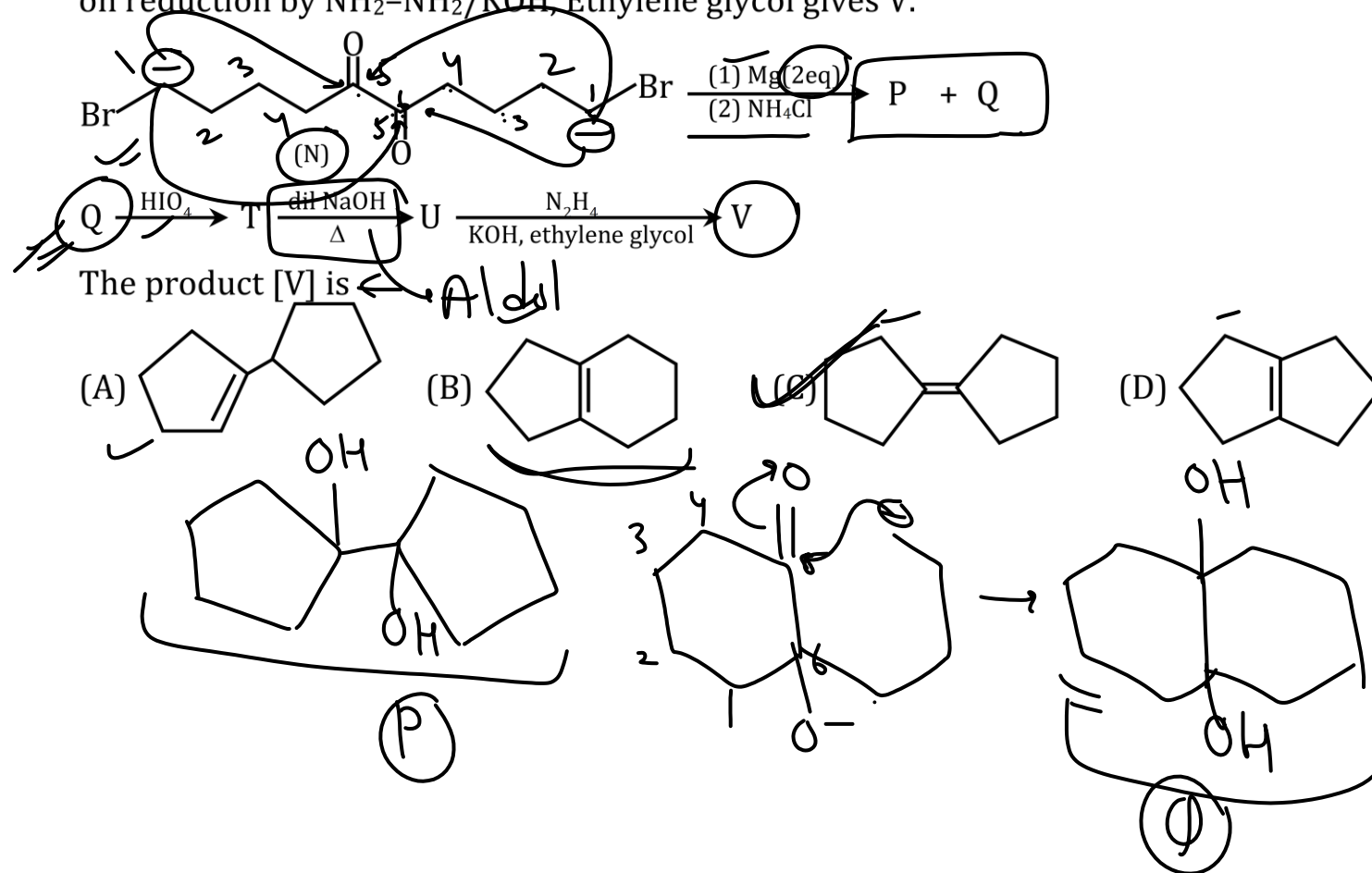
CHEMISTRY

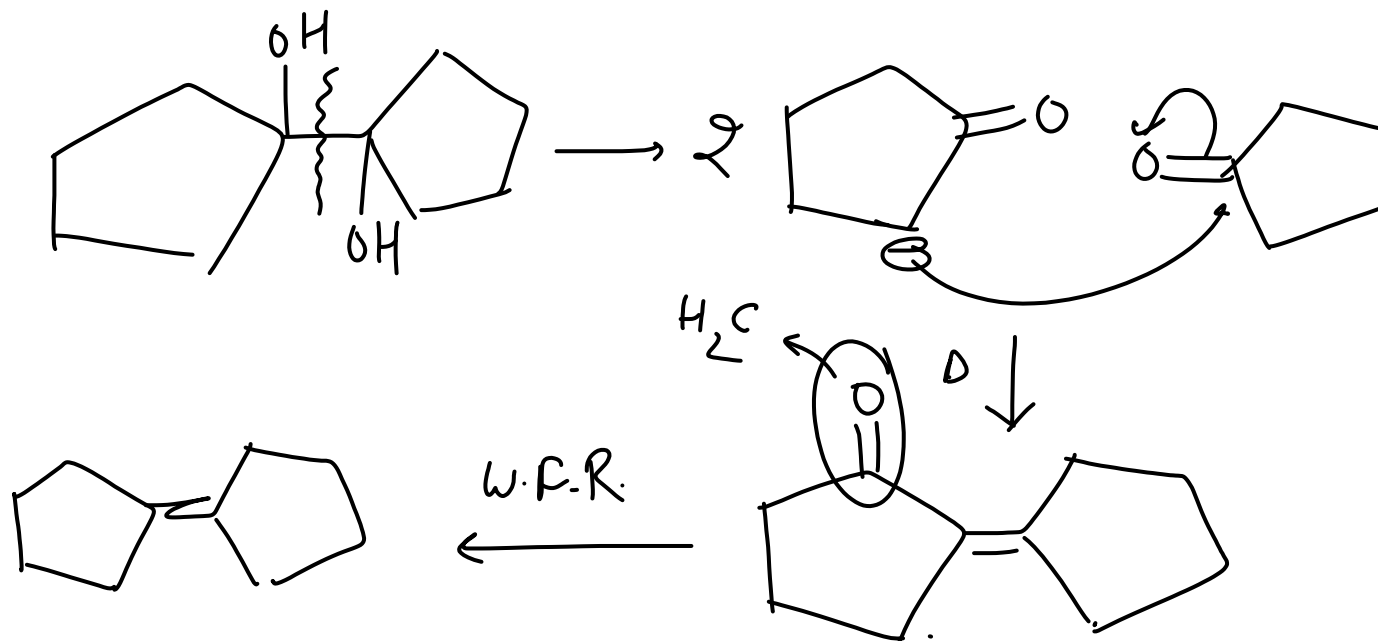
SECTION-I ; (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

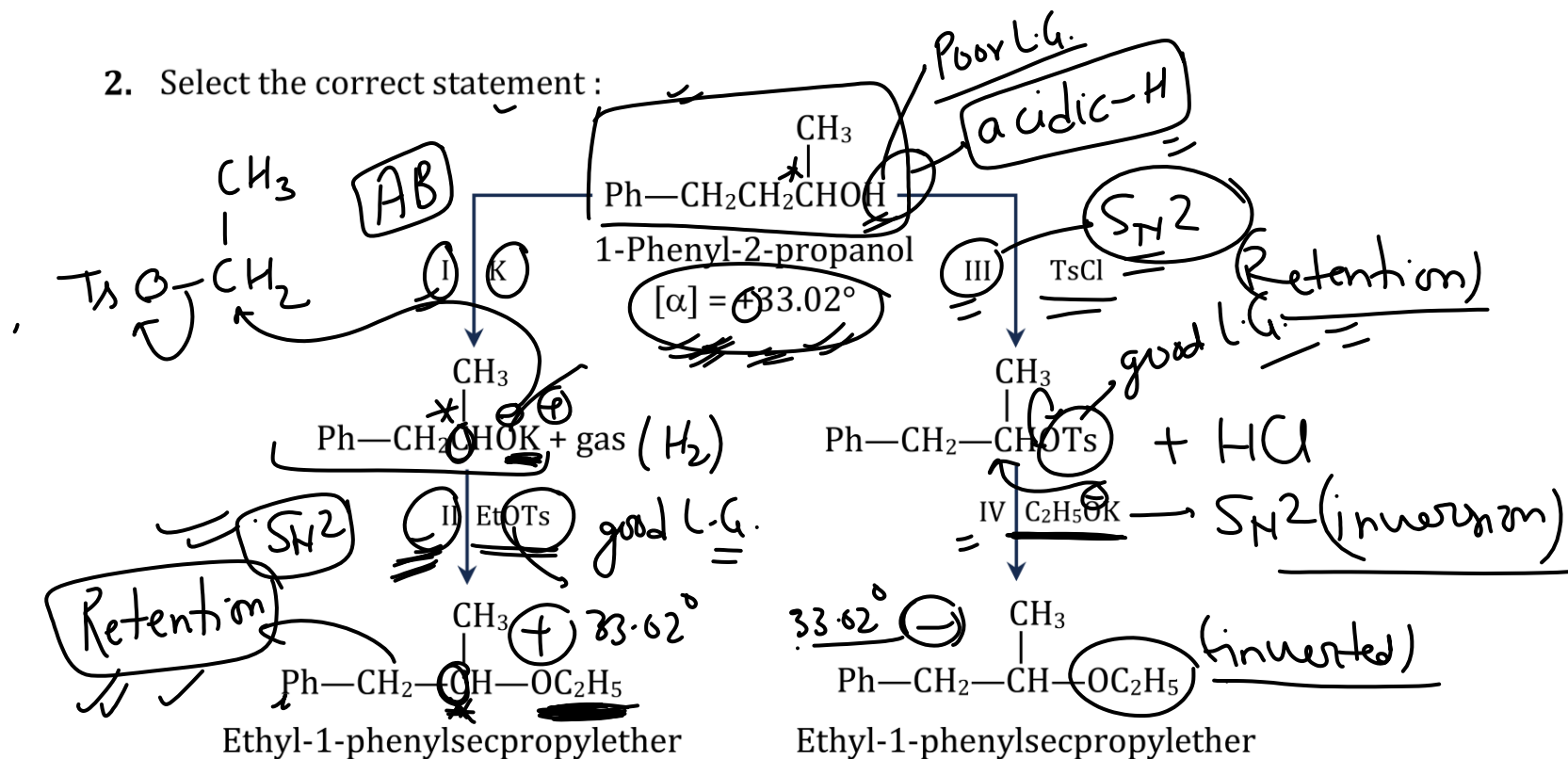
<i>Full Marks</i>	: +3	If ONLY the correct option is chosen;
<i>Zero Marks</i>	: 0	If none of the options is chosen (i.e. the question is unanswered);
<i>Negative Marks</i>	: -1	In all other cases.

1. The reaction of compound N with $\text{Mg}(2\text{eq})$ in $(\text{C}_2\text{H}_5)_2\text{O}$ followed by addition of NH_4Cl gives P and Q are constitutional isomers. The compound P and Q possess same molecular formula. The compound Q on treatment with HIO_4 gives T. The reaction of T with $\text{dil NaOH}/\Delta$ produces U which on reduction by $\text{NH}_2\text{-NH}_2/\text{KOH}$, Ethylene glycol gives V.

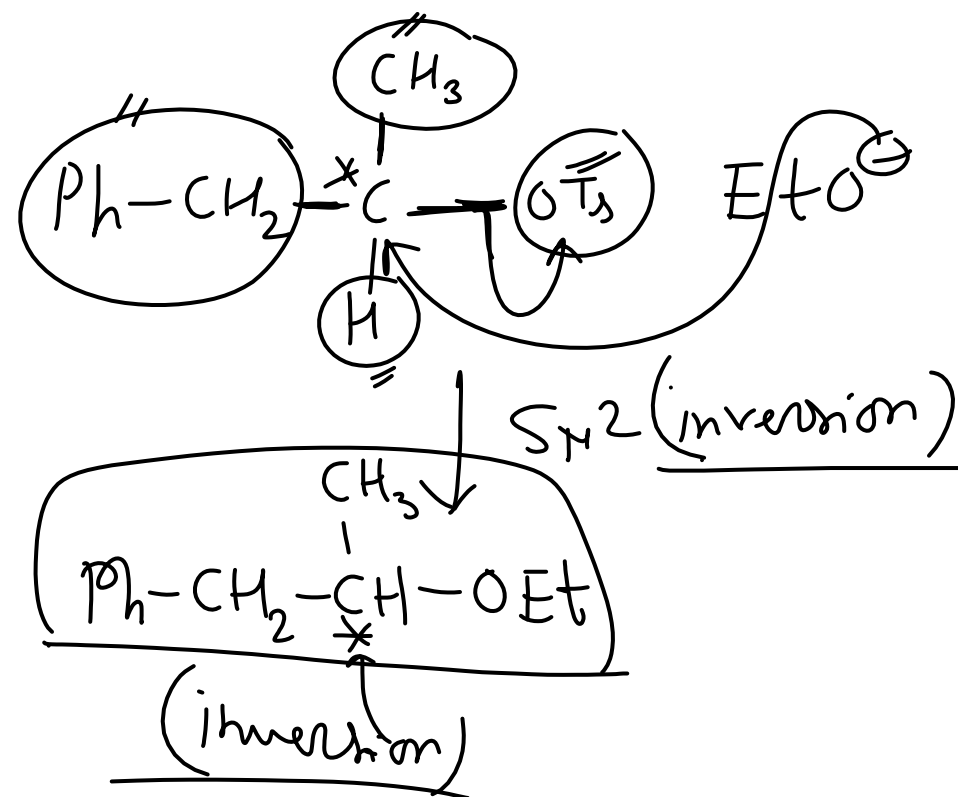




2. Select the correct statement :



- (A) Both the ethers obtained by the two routes have opposite but equal optical rotation. T
- (B) One of the ether is obtained as a racemic mixture. F
- (C) Step II & III both are $\text{S}_\text{N}2$ reaction and both have inversion. F
- (D) Step II has inversion but step III has retention. F



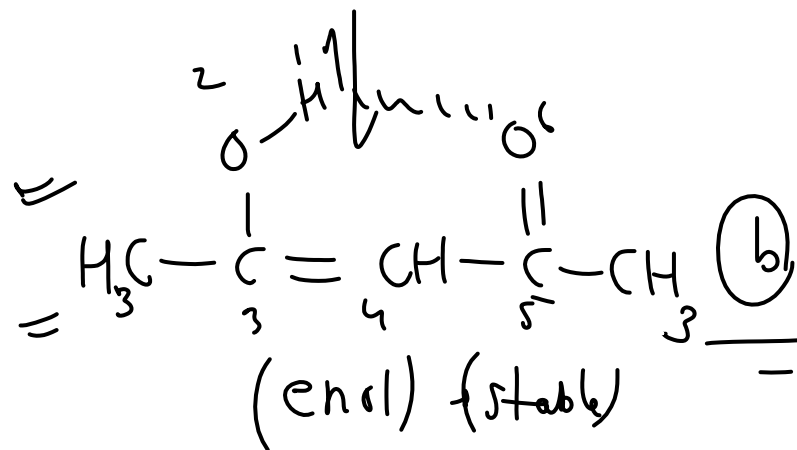
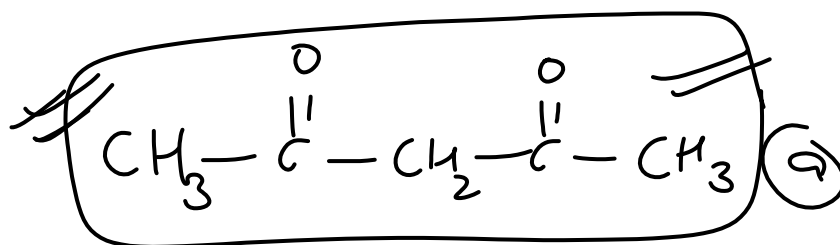
SECTION-II : (Maximum Marks : 32)

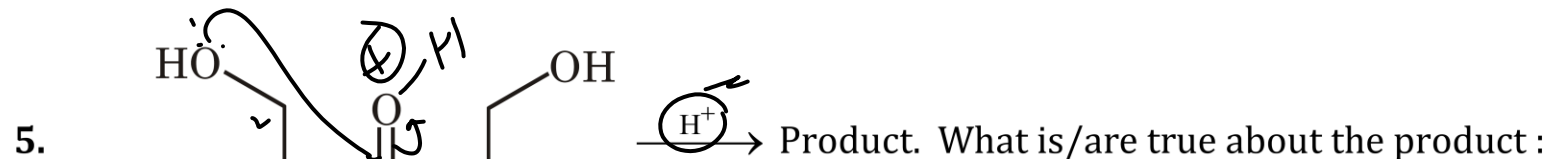
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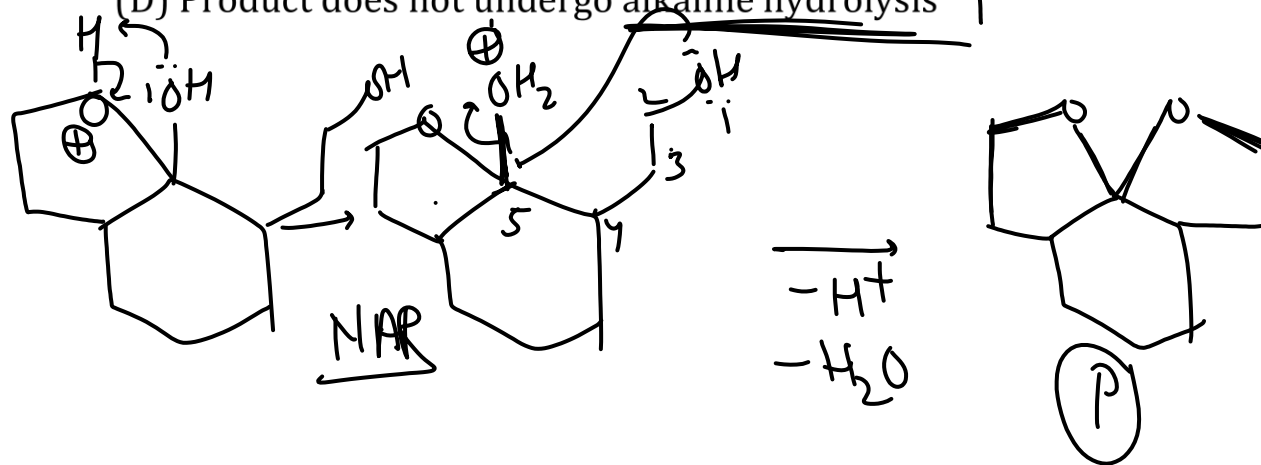
4. In a sample of acetylacetone, which of the following observations is/are True :

- ~~(A)~~ Decolourisation of $\text{Br}_2 / \text{CCl}_4$ solution, if added in acetyl acetone. ~~+~~
~~(B)~~ Appearance of red precipitate, if Brady's reagent is added. ~~+~~
 (C) For acetyl acetone, enol content is less than keto content ~~F~~
 (D) Acetyl acetone decolourise neutral FeCl_3 solution ~~F~~





- (A) Degree of unsaturation in product is 3. T
 (B) Product is formed by nucleophilic addition followed by elimination. T
 (C) Number of oxygen atoms in the product is 2. T
 (D) Product does not undergo alkaline hydrolysis T



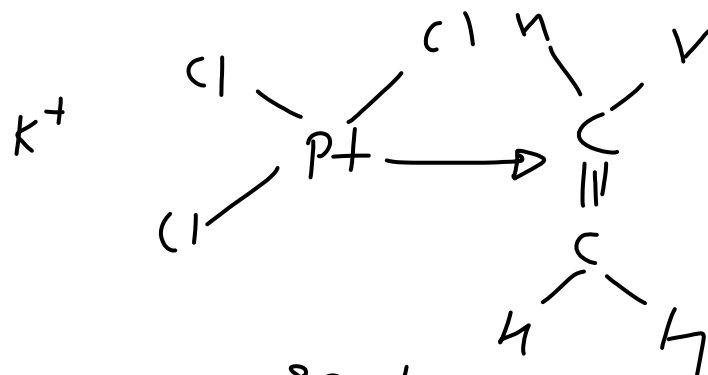
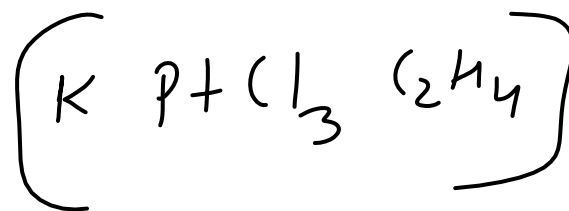
CHEMISTRY

SECTION-I : (Maximum Marks : 18)

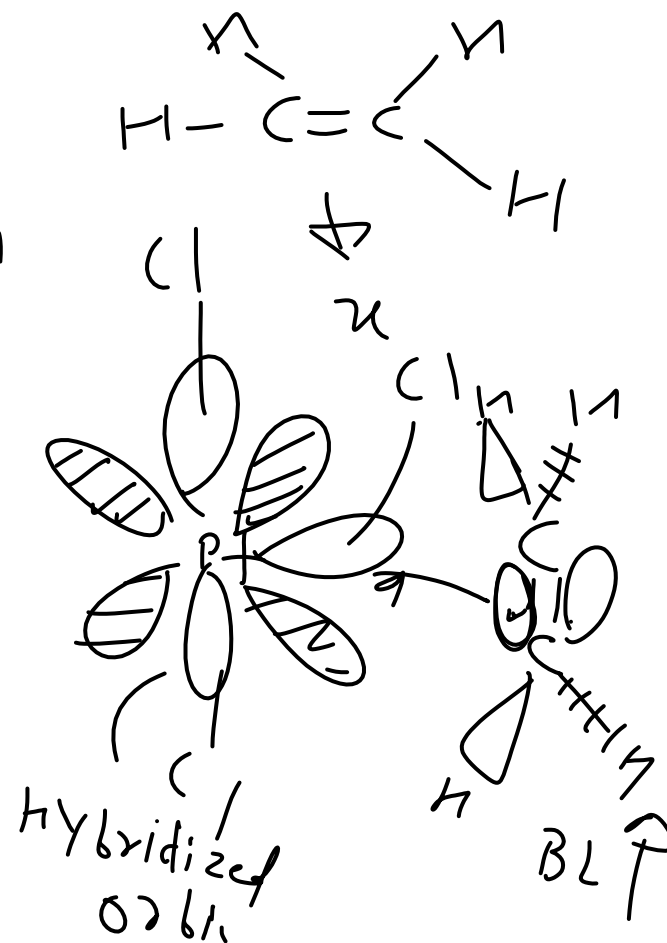
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +3	If ONLY the correct option is chosen;
<i>Zero Marks</i>	: 0	If none of the options is chosen (i.e. the question is unanswered);
<i>Negative Marks</i>	: -1	In all other cases.

3. In isolated condition C - C bond length of C_2H_4 is x, than the bond length of C - C bond of C_2H_4 in Zeise's salt is : _____
- (A) Greater than x (B) Less than x (C) Equal to x (D) None of these



39. planar
dsp²



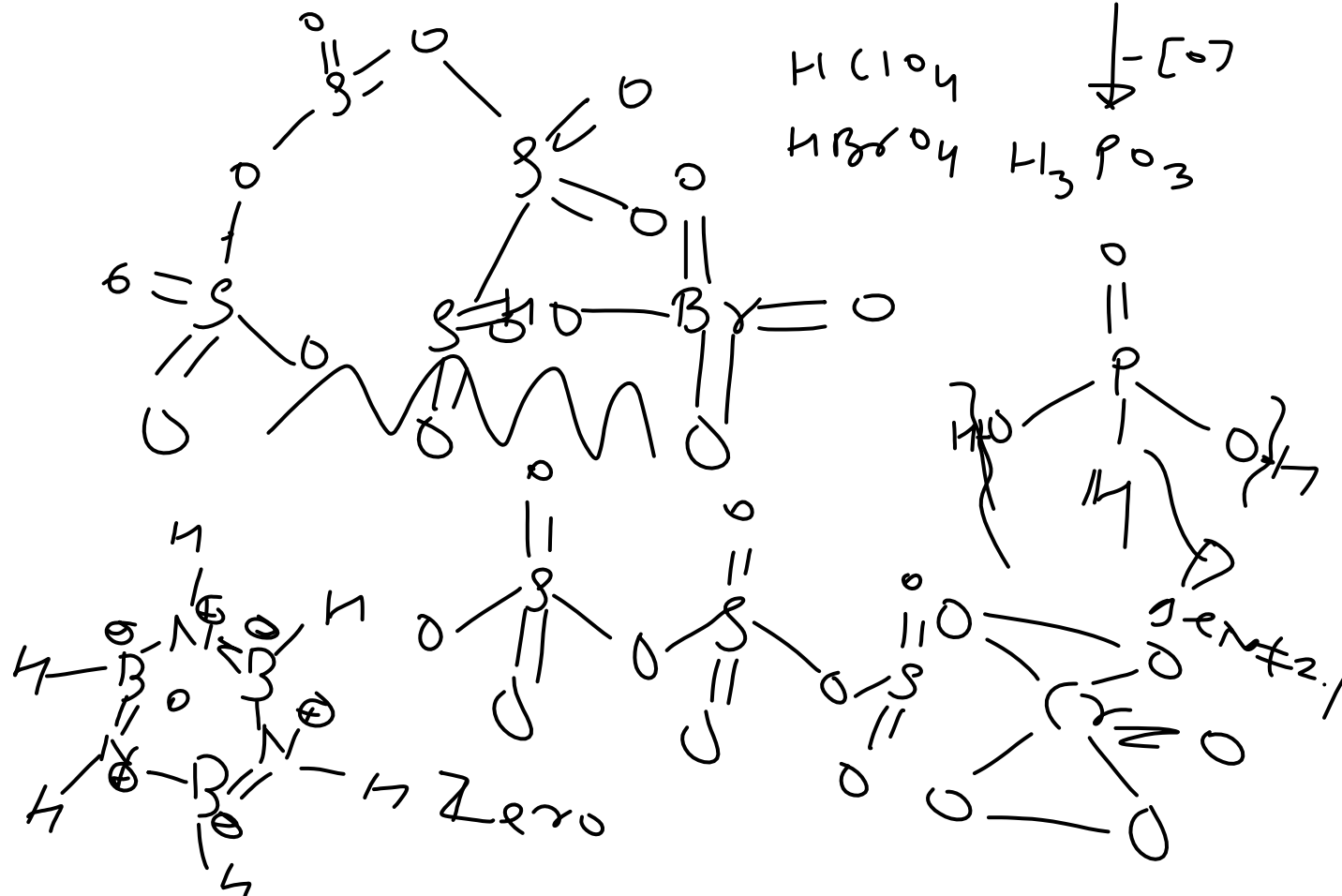
4. Select the correct statement :

(A) Basicity of phosphorous acid is three X

(B) Perbromic acid is having only one peroxy linkage X

(C) β -SO₃ is having cyclic structure X

(D) Borazine is having zero dipole moment



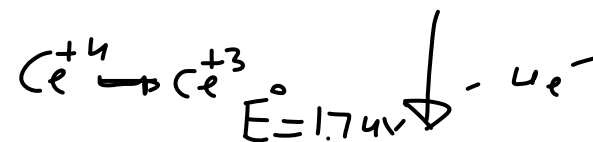
5. Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statement about cerium is incorrect ? _____

(A) The common oxidation states of cerium are +3 and +4 ✓

(B) Cerium (IV) acts as an oxidizing agent ✓

(C) The +4 oxidation state of cerium is more stable in solutions ✗

(D) The +3 oxidation state of cerium is more stable than the +4 oxidation state ✓



6. Which of the following metals can be extracted by both self-reduction and carbon-reduction methods?

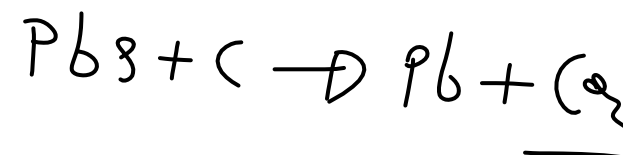
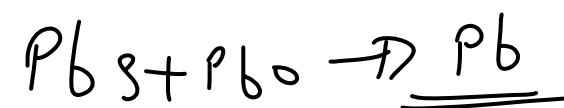
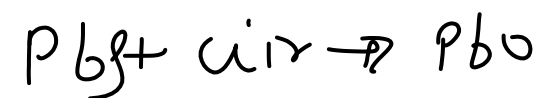
(A) Fe

(B) Al

~~(C) Pb~~

(D) Ag

sulfide ore



SECTION-III: (Maximum Marks: 18)

- This section contains **SIX (06)** question. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

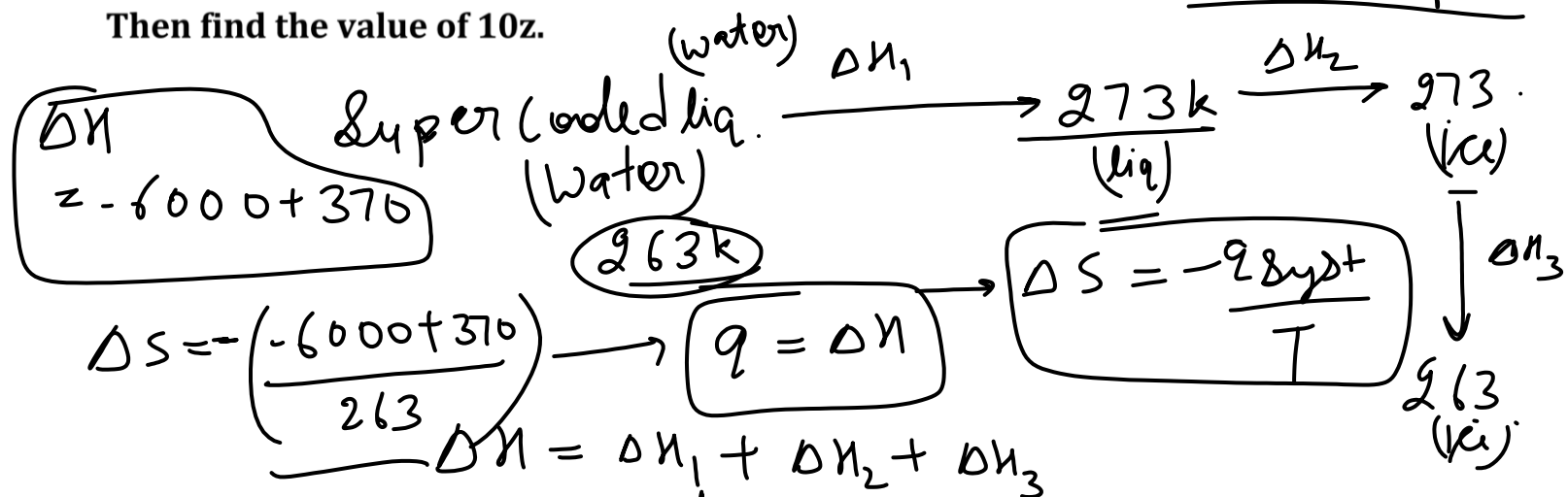
<i>Full Marks</i>	: +3	If ONLY the correct numerical value is entered.
<i>Zero Marks</i>	: 0	In all other cases.

2. The change in entropy of the surroundings (in J K^{-1} , rounded off to two decimal places) is (z) to convert 1 mol of supercooled water at 263 K to ice at 263 K at 1 bar is

[Consider: $\Delta_{\text{fus}}H^\circ$ at $273 \text{ K} = 6.0 \text{ kJ mol}^{-1}$, and the molar heat capacity of water is higher than that of ice by $37.0 \text{ J K}^{-1} \text{ mol}^{-1}$ in the temperature range of 263 K to 273 K]

Then find the value of $10z$.

$(C_p)_{\text{ice}} = a$
 $(C_p)_{\text{water}} = a + 37$



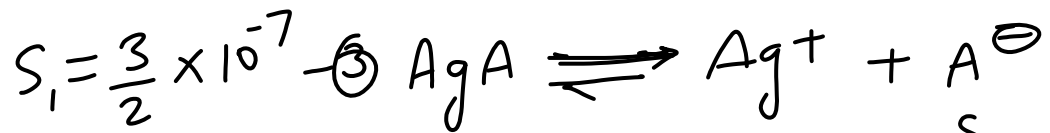
$z = \Delta S = 21.40$
 $\Delta H = \int C_p dT$
 $= (a + 37)10 + (-6 \times 1000) + a(-10)$
 $= 10a + 370 - 6000 + (-10a)$

$10z = 214$

$$\lambda = \frac{\kappa \times 1000}{S}$$

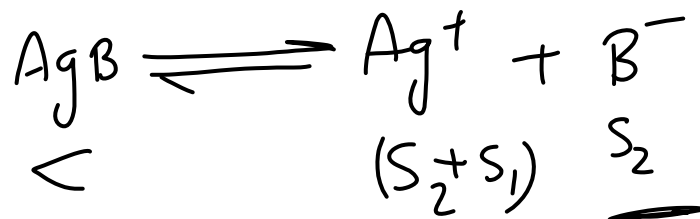
3.

A saturated solution in AgA ($K_{sp} = 3 \times 10^{-14}$) and AgB ($K_{sp} = 1 \times 10^{-14}$) has conductivity of $375 \times 10^{-10} \text{ Scm}^{-1}$ and limiting molar conductivity of Ag^+ and A^- are $60 \text{ Scm}^2 \text{ mol}^{-1}$ and $80 \text{ Scm}^2 \text{ mol}^{-1}$ respectively then what will be the limiting molar conductivity of B^- (in $\text{Scm}^2 \text{ mol}^{-1}$)



$$S_2 = \frac{1}{2} \times 10^{-7} \quad \text{--- (4)}$$

$$\frac{(K_{sp})_{\text{AgA}}}{(K_{sp})_{\text{AgB}}} = \frac{3 \times 10^{-14}}{1 \times 10^{-14}}$$



$$\frac{(S_1 + S_2) S_1}{(S_1 + S_2) S_2} = \frac{3}{S_2} \quad \text{--- (5)}$$

$$(K_{sp})_{\text{AgA}} + (K_{sp})_{\text{AgB}} = (S_1 + S_2) S_1$$

$$\frac{S_1}{S_2} = 3 \quad \text{--- (6)}$$

$$S_1 = 3 S_2$$

$$4 \times 10^{-14} = S_1^2 + S_1 S_2 + S_1 S_2 + S_2^2$$

$$4 \times 10^{-14} = (S_1 + S_2)^2$$

$$S_1 + S_2 = 2 \times 10^{-7} \quad \text{--- (7)}$$

(S₁ + S₂)

$$k_{\text{rel}} = k_{A^+} + k_{A^-} + k_B$$

$$375 \times 10^{-10} = \frac{2 \times 10^{-7} \times 60}{10^3}$$

$$+ \frac{3}{2} \times \frac{10^{-7} \times 40}{1000}$$

$$\times \lambda_B^- \times \frac{1 \times 10^{-7}}{2 \times 10^3}$$

$$375 = 120 + 120 + \frac{\lambda_B^-}{2}$$

(270)

$$\frac{\lambda_B^-}{2} = 375 - 240$$

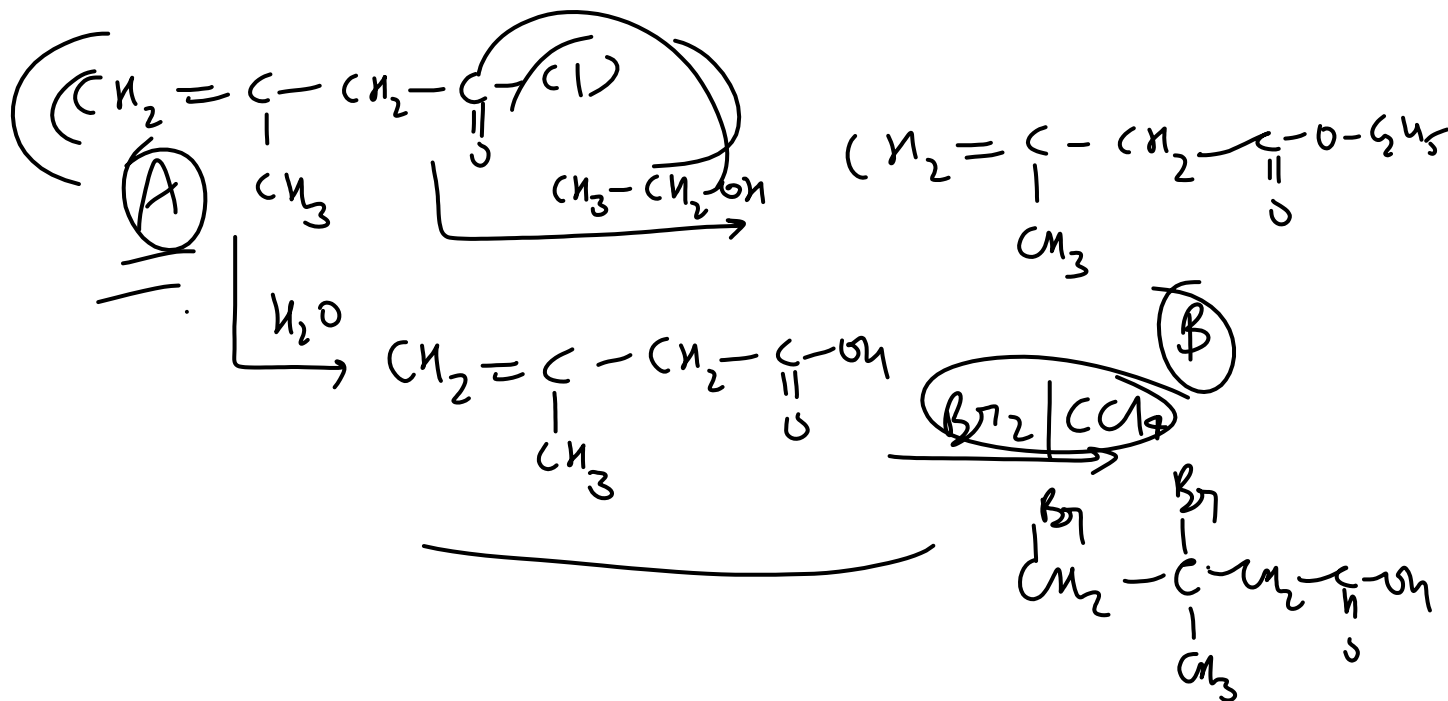
$$\lambda_B^- = 135 \times 2$$

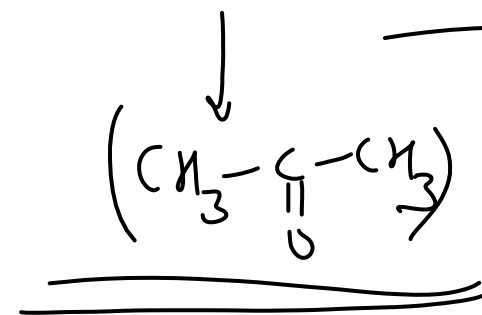
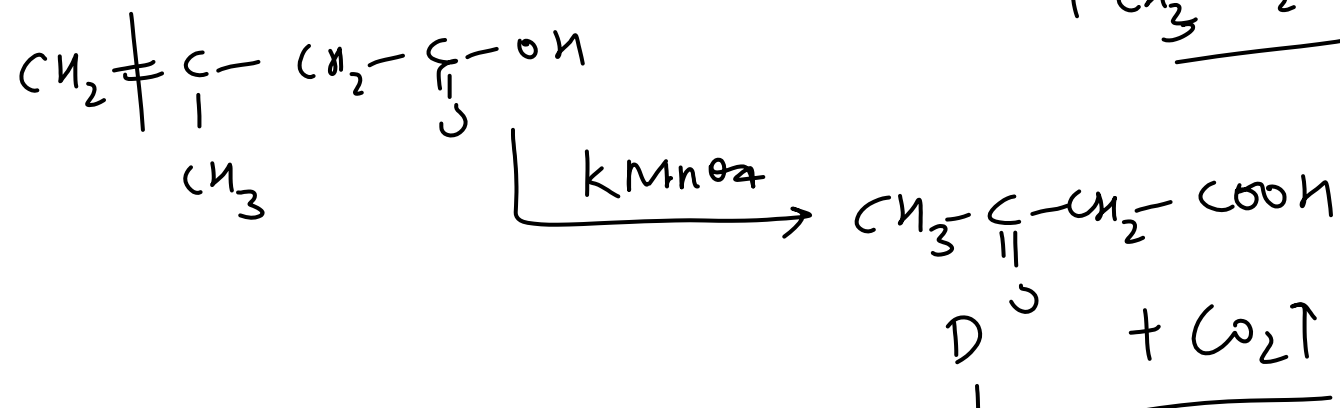
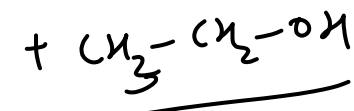
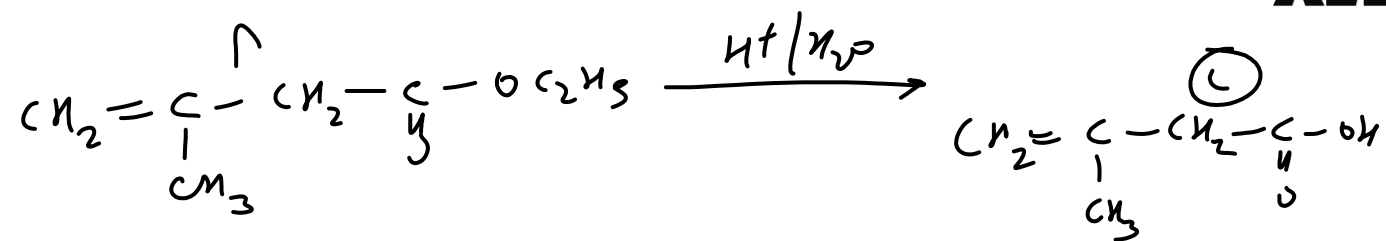
$$= \underline{\underline{270}}$$

$$D.U. = \frac{(5+1)}{2} - \frac{(7+1)}{2} = 2$$

5. An organic compound (A) C_5H_7OCl reacts rapidly with ethanol to give (B) $C_7H_{12}O_2$. (A) also reacts with water to produce acid which reacts with Br_2/CCl_4 to give $C_5H_8Br_2O_2$. 'B' on boiling with aqueous H_2SO_4 formed an acid (C). When C is oxidised with $KMnO_4$ to an acid (D) $C_4H_6O_3$ and CO_2 gas are produced. On mild heating, 'D' gives E. E cannot reduce Tollen's reagent. Calculate the molecular weight of E.

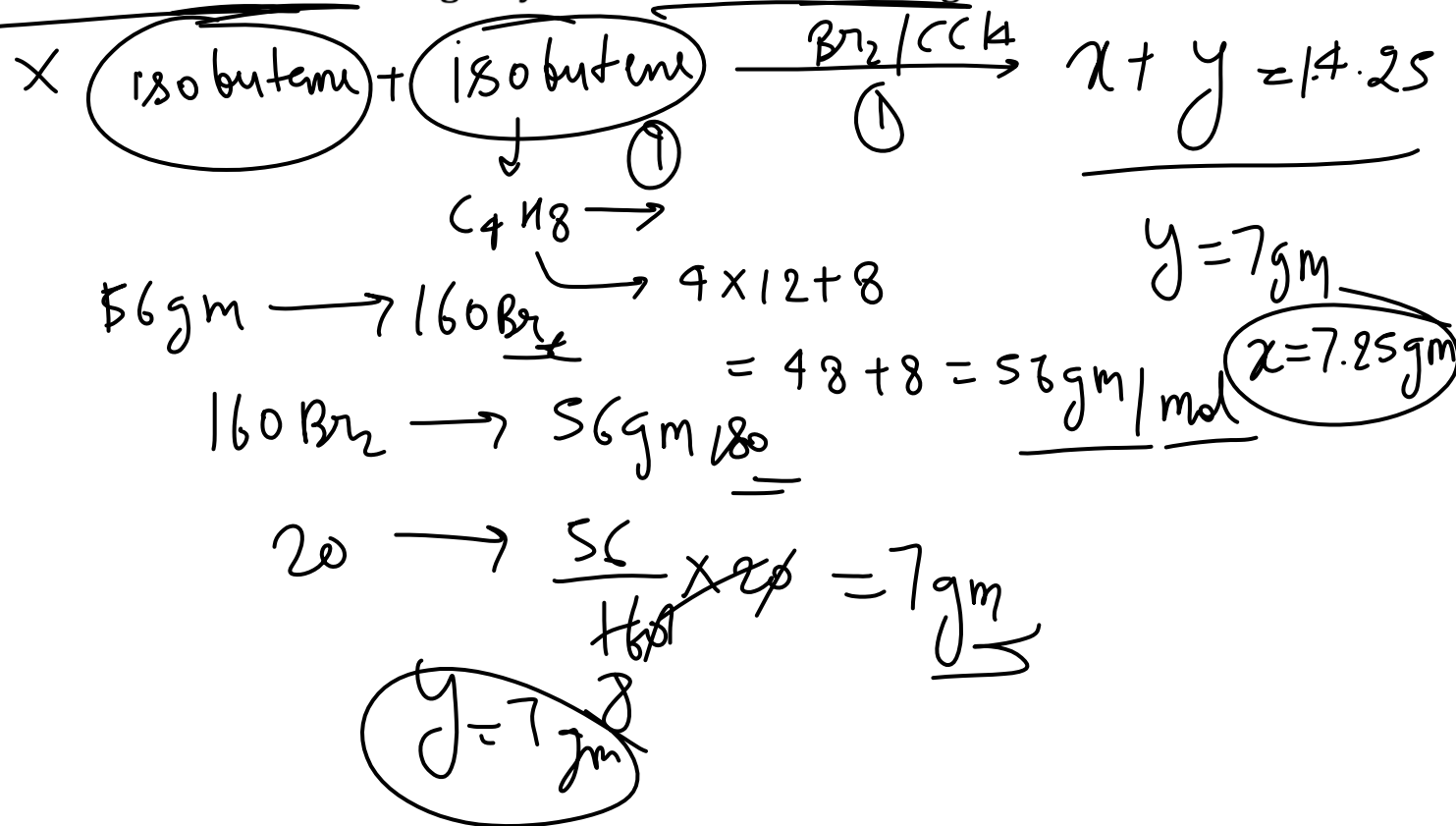
[Give your answer as sum of digits till you get a single digit answer]

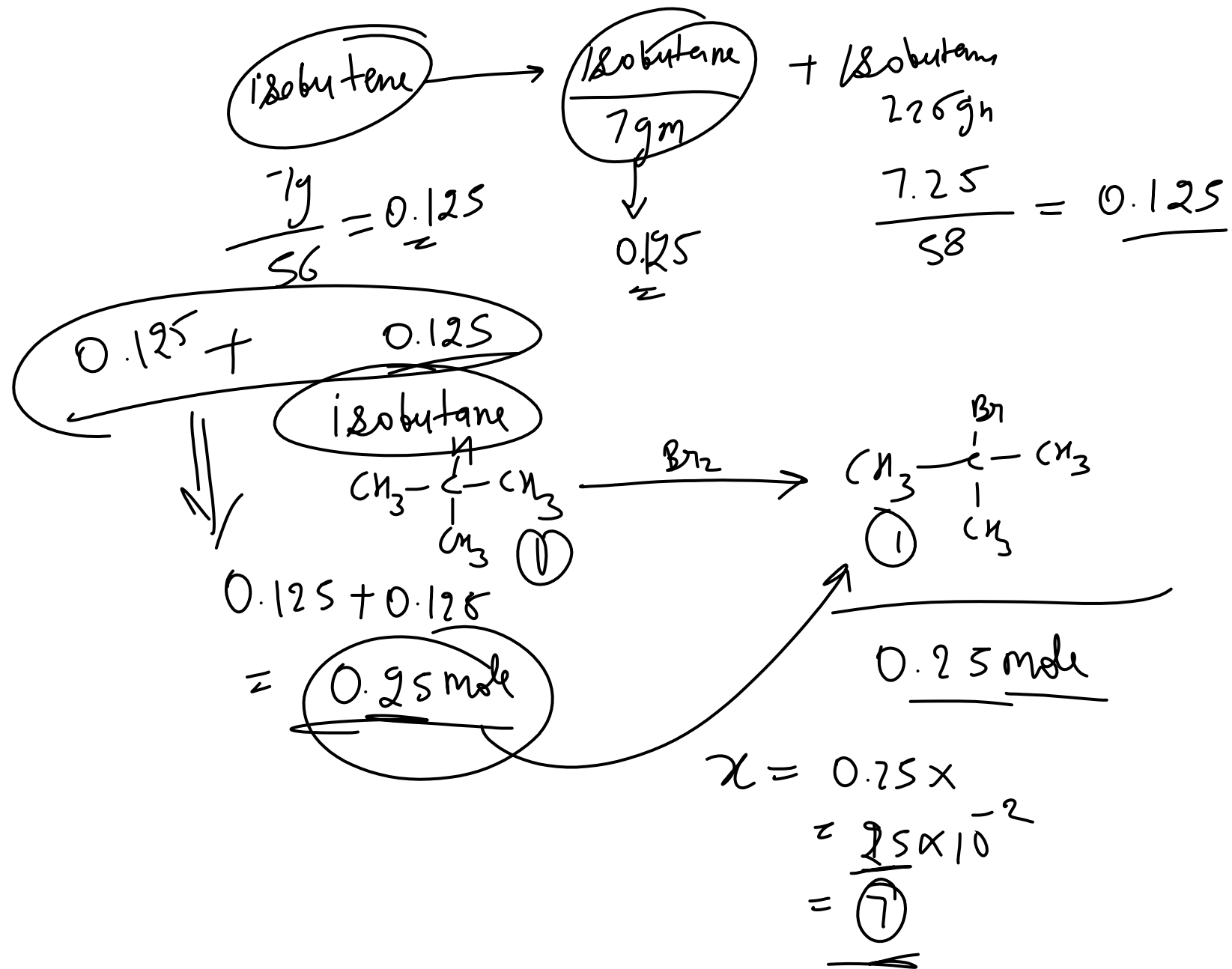




6. A 14.25 g mixture of isobutane and isobutene required 20 gm of Br_2 (in CCl_4) for complete addition. If 14.25 gm of the mixture is catalytically hydrogenated and the entire alkane is monobrominated in the presence of light at 127°C . Moles of major organic product obtained by bromination is (At. wt. of bromine = 80).

→ If your answer is $x \times 10^2$ then give your answer as sum of digits of x



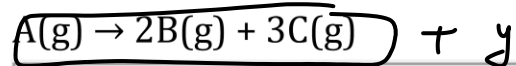


SECTION-II : (Maximum Marks : 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +4	If only (all) the correct option(s) is(are) chosen;
<i>Partial Marks</i>	: +3	If all the four options are correct but ONLY three options are chosen;
<i>Partial Marks</i>	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct;
<i>Partial Marks</i>	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
<i>Zero Marks</i>	: 0	If unanswered;
<i>Negative Marks</i>	: -2	In all other cases.
- For example, if first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For a gaseous first order reaction



Time (sec)	0	90	∞
Total pressure (mm of Hg)	500	1900	<u>2100</u>

Select correct option(s) ($\ln 2 = 0.7$)

~~(A) $t_{3/4} = 60$ sec~~

~~(C) $t_{avg} = 42.86$ sec~~

(B) Pressure of A at $t_{3/4} = 125$ mm Hg $\times$

~~(D) P_{total} at 30 sec = 1300 mm Hg~~

Handwritten solution:

$$A(g) \rightarrow 2B(g) + 3C(g)$$

At $t=0$: $P^\circ + y = 500$ — (1)

At $t=90$: $(P^\circ - x) + 2x + 3x = 1900$ — (2)

At $t=\infty$: $0 + 2P^\circ + 3P^\circ = 2100$ — (3)

From (1) & (3):

$$\begin{aligned} P^\circ + y &= 500 \\ 5P^\circ + y &= 2100 \\ \hline 4P^\circ &= 1600 \\ P^\circ &= 400 \\ y &= 100 \end{aligned}$$

From (2):

$$P^\circ + 4x + y = 1900$$

$$4x = 1400$$

$$x = 350$$

Pressure of A at $t=90$: $P^\circ - x = 400 - 350 = 50$ mm Hg

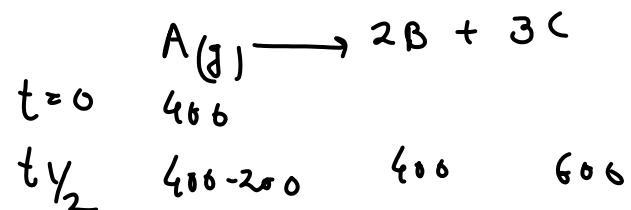
$$K = \frac{1}{t} \ln\left(\frac{P_0}{P_t}\right)_A = \frac{1}{90} \ln \frac{400}{50} = \frac{3 \ln 2}{90} = \frac{\ln 2}{30}$$

$$i) t_{3/4} = \frac{1}{K} \ln \frac{P_0}{P_{3/4}} = \frac{1}{K} \ln 2^3 = \frac{3 \ln 2}{K} = \frac{2 \cdot \ln 2}{\ln 2} \cdot 30 = 60 \text{ sec.}$$

$$ii) \begin{array}{l} P_A^0 \text{ at } t=0 \text{ } 400 \text{ mmHg} \\ \text{at } 3/4 \text{ time} = 400 - 300 = 100 \text{ mmHg} \end{array} \quad \left| \begin{array}{l} t_{1/2} = \frac{1}{K} \ln 2 \\ t_{1/2} = 30 \text{ sec.} \end{array} \right.$$

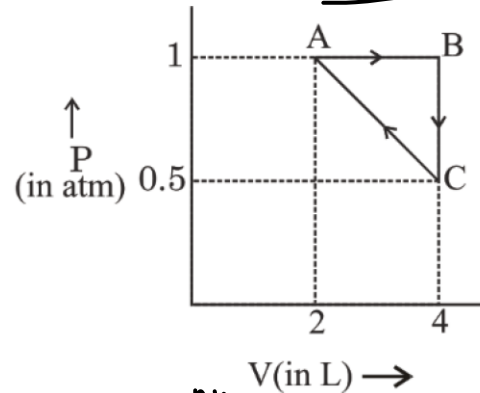
$$iii) t_{avg} = \frac{1}{K} = \frac{t_{1/2}}{\ln 2} = \frac{30}{\ln 2} = \frac{30}{0.7} = \frac{306}{7} = 42.8 \text{ sec.}$$

iv) At $t_{1/2}$



$$\begin{aligned} P_{\text{Total}} &= P_A + P_B + P_C + P \\ &= 200 + 400 + 600 + 100 = 1300 \text{ mmHg} \end{aligned}$$

2. An ideal gas undergoes the following P-V process, Select the correct statements
(Given 1 atm-L = 101.325 J)



A) For cyclic process, change in state function = 0

$$\Delta H, \Delta U, \Delta S = 0$$

$$\Delta P, \Delta T, \Delta V$$

- ☒ (A) The enthalpy change for overall process is zero.
☒ (B) For the overall process $\Delta H = \Delta U = 0$
☒ (C) Heat exchanged in cyclic process (ABCA) is 50.66 J.
☒ (D) Total work = - 50.66 J

FL07

$$\Delta U = Q + W$$

$$\therefore \Delta U = 0$$

$$Q = -W$$

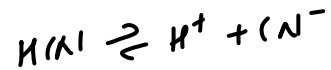
$$Q = -W$$

$$= -(-50.66)$$

$$= 50.66 \text{ J}$$

$$\text{Work} = \text{Area of closed structure} \\ = \frac{1}{2} \times 2 \times \frac{1}{2} = \frac{1}{2} \text{ atm-L}$$

$$W = -0.5 \text{ atm-L} \\ = -\frac{1}{2} \times 101.325 = -50.66 \text{ J}$$



3. Calculate the solubility of AgCN in a buffer solution of pH = 3. Given K_{sp} of AgCN = 1.2×10^{-16} and

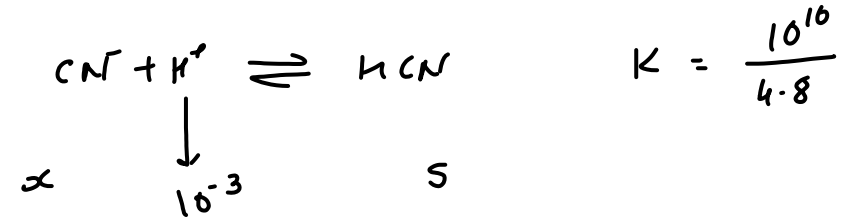
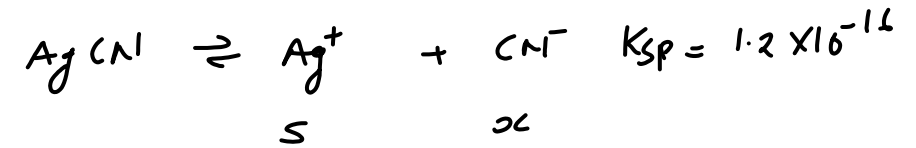
K_a for HCN = 4.8×10^{-10} .

(A) 2.55×10^{-5} mol litre⁻¹

(B) 1.58×10^{-5} mol litre⁻¹

(C) 4.57×10^{-5} mol litre⁻¹

(D) 5.60×10^{-5} mol litre⁻¹



$$K_{sp} = [Ag^+][CN^-] = s \cdot x = 1.2 \times 10^{-16} \quad \text{--- (1)}$$

$$K = \frac{[HCN]}{[CN^-][H^+]} = \frac{s}{x \cdot 10^{-3}} = \frac{10^{10}}{4.8} \quad \text{--- (2)}$$

$$s \cdot x \cdot \frac{s}{x \cdot 10^{-3}} = 1.2 \times 10^{-16} \times \frac{10^{10}}{4.8}$$

$$s^2 = \frac{10^{-9}}{4}$$

$$s = \sqrt{\frac{10^{-9}}{4}} = \sqrt{\frac{25}{10} \times 10^{-10}} = \frac{5 \times 10^{-5}}{\sqrt{10}}$$

SECTION-III: (Maximum Marks: 18)

- This section contains **SIX (06)** question. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +3	If ONLY the correct numerical value is entered.
<i>Zero Marks</i>	: 0	In all other cases.

1. The distance between the two radial nodes is d Å for the given orbital of H-atom whose wave function is given as

(35) $\psi = \frac{1}{9\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (\sigma^2 - 4\sigma + 3) e^{-\sigma/2}$ at Radial Node: ψ & $\psi^2 = 0$

where a and z are constants and $\sigma = \frac{Zr}{n.a_0}$, find value of 'd'?

$$0 = \sigma^2 - 4\sigma + 3$$

$$\sigma = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 3}}{2}$$

$$\sigma = \frac{4 \pm \sqrt{4}}{2}$$

$$\sigma_1 = \frac{4 - \sqrt{4}}{2} = 1$$

$$\sigma_2 = \frac{4 + \sqrt{4}}{2} = 3$$

$$\sigma^0 (\sigma^2 - 4\sigma + 3)$$

$$l = 0$$

$$n - l - 1 = 2$$

$$n = 3$$

Hence orbital is 3s.

$$r_1 = \frac{n\sigma_1 a_0}{Z} = \frac{3 \times 1 \times a_0}{2 \times 1} = \frac{3a_0}{2}$$

$$r_2 = \frac{n\sigma_2 a_0}{Z} = \frac{3 \times 3 \times a_0}{2 \times 1} = \frac{9a_0}{2}$$

Distance b/w 2 radial Nodes

$$\frac{9a_0}{2} - \frac{3a_0}{2} = \frac{6a_0}{2} = 3a_0 = 3 \times 0.529 \text{ Å}$$

$$= \underline{\underline{1.58 \text{ Å}}}$$

JEE

ADVANCED 2024



Mathematics

FINAL PRACTICE
PAPER-2



HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

MATHEMATICS

SECTION-I : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

$$y^2 = 5x \geq 0$$

1. Let two conics be given by P: $y^2 = 5x$ and H: $x^2 - y^2 = 6$, P and H intersect at points A and B in first and fourth quadrant respectively. Tangents to P and H at point 'A' intersect the x-axis at points L and M respectively then ratio of areas of $\triangle LAB$ and $\triangle MAB$ is

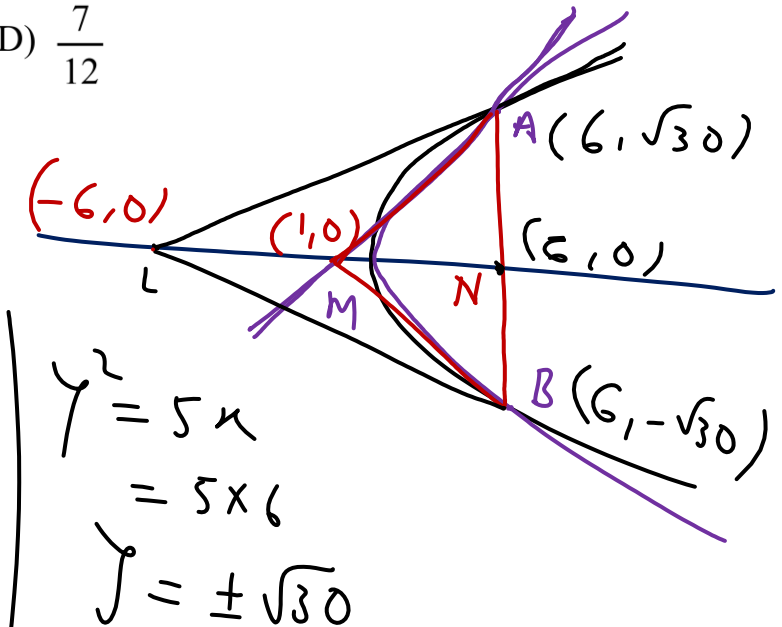
$$\frac{\triangle LAB}{\triangle MAB}$$

$$= \frac{\frac{1}{2} AB \times LN}{\frac{1}{2} AB \times MN} = \frac{12}{5}$$

(A) $\frac{12}{5}$
(C) $\frac{12}{7}$

(B) $\frac{5}{12}$
(D) $\frac{7}{12}$

$$\begin{aligned} y^2 &= 5x \quad \& \quad x^2 - y^2 = 6 \\ x^2 - 5x - 6 &= 0 \\ (x-6)(x+1) &= 0 \\ x &= 6 \quad (x \neq -1) \end{aligned}$$



$$y^2 = 5x$$

Tangent at $A(6, \sqrt{30})$

$$y(\sqrt{30}) = 5\left(\frac{x+6}{2}\right)$$

intersects x axis at $y=0$

$$\therefore \boxed{x = -6} \quad \therefore L(-6, 0)$$

Tangent to

$$x^2 - y^2 = 6$$

at $A(6, \sqrt{30})$

$$T = 0$$

$$x \cdot 6 - y \cdot \sqrt{30} = 6$$

intersects $y=0$

$$\text{at } x = \frac{6}{6} = 1$$

⊕
500 | 500 ✓

2. An exam consist of 11 problems selected randomly from a collection of 1000 different problems. For a student to pass, he needs to solve correctly at least six of the 11 problems . if the students knows to solve exactly half of 1000 problems, then the probability that the student pass the exam is

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{1}{5}$

✓ (D) $\frac{1}{2}$

Let $P(A_i)$ = The prob. when student attempts exactly 'i' questions correctly.

$$P(A_0) = P(A_{11}) ; P(A_1) = P(A_{10})$$

$$P(A_2) = P(A_9) \quad \dots$$

$$P_{\text{req}} = P(A_6) + P(A_7) + P(A_8) + P(A_9) + P(A_{10}) \\ + P(A_{11})$$

$$= \frac{1}{2} \left[2 P(A_6) + 2 P(A_7) + \dots + 2 P(A_{11}) \right]$$

$$= \frac{1}{2} \left[P(A_0) + P(A_1) + P(A_2) + \dots + P(A_9) + P(A_{10}) + P(A_{11}) \right]$$

$$= \frac{1}{2} \times 1 = \frac{1}{2}$$



$$z^3 + 1 = 0 \Rightarrow z \neq z_1, z_2, z_3, \dots, z_{10}$$

3. Let $z_1, z_2, z_3, \dots, z_{10}$ are roots of equation $z^{12} + z^{10} + z^9 + z^8 + z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + 1 = 0$ such that $\boxed{z_r^3 + 1 \neq 0} \forall r \in \{1, 2, 3, \dots, 10\}$, then value of $(z_1^{2021} + z_2^{2021} + \dots + z_{10}^{2021})$ is-

(A) 4

(B) 5

☒ (C) -1

(D) 0

$$\Rightarrow (z^{12} - z) + (z^{10} + z^9 + z^8 + \dots + z^2 + z + 1) = 0$$

$$\Rightarrow z(z^{11} - 1) + \frac{1 \cdot (z^{11} - 1)}{z - 1} = 0 \Rightarrow (z^{11} - 1) \frac{(z^2 - z + 1)}{z - 1} = 0$$

$$\Rightarrow (z^{11} - 1) \left(z + \frac{1}{z - 1} \right) = 0$$

$$\left\{ \begin{array}{l} z^3 + 1 \neq 0 \\ (z + 1)(z^2 - z + 1) \neq 0 \end{array} \right\}$$

$$\Rightarrow z^{11} - 1 = 0$$

$\therefore z$ is 11^{th} root of unity.

$\therefore$ Roots are $1, z_1, z_2, z_3, \dots, z_{10}$

For these roots

$$1^k + z_1^k + z_2^k + \dots + z_{10}^k = \begin{cases} 0 & \text{if } k \text{ is NOT a multiple of } 11 \\ 11 & \text{otherwise} \end{cases}$$

$$\therefore 1^{2021} + z_1^{2021} + z_2^{2021} + \dots + z_{10}^{2021} = 0$$

$$z_1^{2021} + z_2^{2021} + \dots + z_{10}^{2021} = -1$$

4. Let $A = [a_{ij}]_{n \times n}$ be a square matrix such that $a_{ij} = (3i + 2)5^{-j}$. then the value of $\lim_{n \rightarrow \infty} (\text{trace}(A^n))^{1/n}$ is,
($n \in \mathbb{N}$)

(A) $\frac{23}{16}$

(B) $\frac{16}{23}$

(C) $\frac{23}{32}$

(D) $\frac{32}{23}$

$$A = \begin{bmatrix} 5 & 0 & 0 & \dots & 0 \\ 8 & 0 & 0 & \dots & 0 \\ 11 & 0 & 0 & \dots & 0 \\ \vdots & 0 & 0 & \dots & 0 \\ 3n+2 & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-3} & \dots & 5^{-n} \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

$$A = PQ$$

$$A^2 = (PQ)^2 = \underbrace{PQ}_{PQ} PQ$$

$$\begin{bmatrix} 5^{-1} & 5^{-2} & 5^{-3} & \dots \\ 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \\ 0 & 0 & 0 & \dots \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 & \dots \\ 8 & 0 & 0 & \dots \\ 11 & 0 & 0 & \dots \\ \vdots & 0 & 0 & \dots \\ 3n+2 & 0 & 0 & \dots \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 0 & 0 & \dots & 0 \\ 8 & 0 & 0 & \dots & 0 \\ 11 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 3i+2 & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \sum_{i=j=1}^n (3i+2) 5^{-j} & 0 & 0 & \dots & 0 \\ 0 & \dots & \dots & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 \end{bmatrix} \begin{bmatrix} 5^1 & 5^2 & \dots & 5^n \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

$$A^2 = (\text{tr } A) \cdot A$$

$$A^3 = (\text{tr } A) A^2 = (\text{tr } A) (\text{tr } A) A$$

$$\therefore A^3 = (\text{tr } A)^2 A \Rightarrow A^n = (\text{tr } A)^{n-1} A$$

$$(\text{tr } A)^{n-1} \begin{bmatrix} a_{11} & & \\ & a_{22} & \\ & & \ddots \\ & & & a_{nn} \end{bmatrix}$$

$$\therefore \lim_{n \rightarrow \infty} \left[(\text{tr } A^n) \right]^{1/n} = \lim_{n \rightarrow \infty} \left\{ (\text{tr } A)^n \right\}^{1/n} \quad \because \text{tr } (A^n) = (\text{tr } A)^n$$

$$= \lim_{n \rightarrow \infty} \sum_{i=j=1}^n (3i+2) 5^{-j}$$

$$= 5 \cdot \frac{1}{5} + 8 \cdot \frac{1}{5^2} + 11 \cdot \frac{1}{5^3} + \frac{14}{5^4} + \dots \infty$$

$$= \frac{1}{5} \left[5 + \frac{8}{5} + \frac{11}{5^2} + \frac{14}{5^3} + \dots \infty \right]$$

$$= \frac{1}{5} \left[\frac{5}{1 - \frac{1}{5}} + \frac{3 \cdot \frac{1}{5}}{\left(1 - \frac{1}{5}\right)^2} \right] = \frac{1}{5} \left(\frac{25}{4} + \frac{3}{8} \cdot \frac{8}{4 \cdot 4} \right)$$

$$= \frac{1}{5} \left(\frac{100+15}{16} \right) = \frac{23}{16}$$

5. The value of $\log_{10} \left(\frac{\sum_{i=0}^r \binom{n}{2i} \binom{n-2i}{r-i}}{\sum_{i=r}^n \binom{n}{i} 2^i \binom{2r}{2r-i} \left(\frac{3}{4}\right)^{n-i} \left(\frac{1}{4}\right)^{i-r}} \right)$ ($n \geq 2r$) is :

(A) 2

(B) 1

(C) 0

(D) None

$$\frac{n!}{(n-2i)!(2i)!} \times \frac{(n-2i)!}{(r-i)!(n-r-i)!} \cdot \frac{(r+i)!}{(r+i)!}$$

$$\sum_{i=0}^r \binom{n}{2i} \binom{n-2i}{r-i}$$

$$\Rightarrow \binom{n}{r} \cdot \binom{r}{r} + \binom{n}{r+1} \cdot \binom{r+1}{r-1} + \binom{n}{r+2} \cdot \binom{r+2}{r-2} + \dots + \binom{n}{2r} \cdot \binom{2r}{0}$$

$$\left(\underline{\underline{1+x+x^2}} \right)^n = \underline{\underline{(1+x(1+x))^n}} \Leftarrow \text{coefficient of } x^{2r}$$

$$\sum_{p=0}^n {}^nC_p \cdot \underline{x^p} \cdot \underline{(1+x)^p}$$

$$\sum_{p=0}^n {}^nC_p \cdot {}^{p \cdot} C_{2r-p} \quad \underline{x^{2r-p}}_{2r-(r+1)}$$

$${}^nC_r \cdot {}^r C_r + {}^nC_{r+1} \cdot {}^{r+1} C_{r-1} + \dots + {}^nC_{2r} \cdot {}^{2r} C_0$$

$$\left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right)^n \Rightarrow \text{coeff of } x^{2r}$$

$$\sum_{i=0}^n \underline{n C_i} \cdot \left(\frac{3}{4}\right)^{n-i} \cdot \left(x + \frac{1}{2}\right)^{2i}$$

$$\text{Coeff of } x^{2r} \Rightarrow \sum_{i=0}^n n C_i \left(\frac{3}{4}\right)^{n-i} \cdot {}^{2i}C_{2r} \left(\frac{1}{2}\right)^{2i-2r}$$

$$= \left(\sum_{i=0}^n \underline{n C_i} \cdot \underline{{}^{2i}C_{2r}} \left(\frac{3}{4}\right)^{n-i} \cdot \left(\frac{1}{4}\right)^{i-r} \right)$$

6. A continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the functional equation $f(4x) - 2f(2x) + f(x) = x^2$ such that

$f(0) = 1$

Then the value of $f(3\sqrt{3})$ is

(A) 1

(B) 2

(C) 3

(D) 4

$$\underline{f(4x) - 2f(2x) + f(x) = x^2}$$

$$f(x) = ax^2 + bx + 1$$

$$(16ax^2 + 4bx + 1) - 2(4ax^2 + 2bx + 1) + (ax^2 + bx + 1)$$

$$9ax^2 + bx = x^2$$

$$\boxed{9a = 1}, b = 0$$

$$f(x) = \frac{1}{9} x^2 + 1$$

$$f(3\sqrt{3}) = 4$$

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let A and B be two invertible matrices of order n such that $I - AB$ is invertible, then which of the following is (are) CORRECT?

(A) $I - BA$ is singular

~~(B) $I - BA$ is non-singular~~

~~(C) $(I - BA)^{-1} = I + B(I - AB)^{-1}A$~~

~~(D) $(I - BA)^{-1} = I - B(I - AB)^{-1}A$~~

$$|A| \neq 0, |B| \neq 0 \quad |I - AB| \neq 0$$

$$I - BA = BB^{-1} - BA$$

$$= B(B^{-1} - A)$$

$$= B(I B^{-1} - A B B^{-1})$$

$$|I - BA| = |B| |I - AB| |B^{-1}| \neq 0$$

$$(I - BA)^{-1} = \left\{ B (I - AB) B^{-1} \right\}^{-1}$$

$$(I - BA)^{-1} = B (I - AB)^{-1} B^{-1}$$

$$= B \left(\underline{I - AB} \right)^{-1} \left(\underline{(I - AB) + AB} \right) B^{-1}$$

$$= \underline{B} \left(\underline{I} + \underline{(I - AB)^{-1} AB} \right) \underline{B^{-1}}$$

$$= I + B (I - AB)^{-1} \underbrace{AB B^{-1}}$$

$$= I + B (I - AB)^{-1} A$$

2. For distinct complex numbers $z_1, z_2, \dots, z_n$, ($n > 2$) the value of

$$\frac{|z_2 - z_1|^2 + |z_3 - z_2|^2 + \dots + |z_n - z_{n-1}|^2}{|z_n - z_1|^2}$$

can not be less than

(A) $\frac{1}{n-1}$

(C) $\frac{1}{n+1}$

(B) $\frac{1}{n}$

(D) $\frac{1}{n+2}$ $\frac{1}{n-1} > \frac{1}{n} > \frac{1}{n+1} > \frac{1}{n+2}$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

$$z_n - z_1 = (z_n - z_{n-1}) + (z_{n-1} - z_{n-2}) + \dots + (z_2 - z_1)$$

$$|z_n - z_1| = |(z_n - z_{n-1}) + (z_{n-1} - z_{n-2}) + \dots + (z_2 - z_1)|$$

$$|z_n - z_1| \leq |z_n - z_{n-1}| + |z_{n-1} - z_{n-2}| + \dots + |z_2 - z_1|$$

$$RMS \geq AM$$

$$\sqrt{\frac{a^2+b^2+c^2}{3}} \geq \frac{a+b+c}{3}$$

$$\sqrt{\frac{|z_n - z_{n-1}|^2 + |z_{n-1} - z_{n-2}|^2 + \dots + |z_2 - z_1|^2}{n-1}} \geq \frac{|z_n - z_{n-1}| + \dots + |z_2 - z_1|}{n-1}$$

$$\frac{|z_n - z_{n-1}|^2 + |z_{n-1} - z_{n-2}|^2 + \dots + |z_2 - z_1|^2}{(n-1)} \geq \frac{|z_n - z_1|^2}{(n-1)^2}$$

$$\frac{|z_n - z_{n-1}|^2 + |z_{n-1} - z_{n-2}|^2 + \dots + |z_2 - z_1|^2}{|z_n - z_1|^2} \geq \frac{1}{n-1} > \frac{1}{n} > \frac{1}{n+1} > \frac{1}{n+2}$$

3. Let $\int_0^\infty e^{-x^2} dx = a$ and $I_n = \int_0^\infty x^n e^{-x^2} dx, n \in \mathbb{N}$, then

(A) $I_n = \left(\frac{n-1}{2}\right) I_{n-2}$

(C) $I_n = \frac{1}{2} \left(\frac{n-2}{2}\right)! a$, if n is even

(B) $I_n = \frac{(n-1)!}{2^{n-1} \left(\frac{n-2}{2}\right)!} a$, if n is even

(D) $I_n = \frac{(n-1)!}{2^{n-1} \left(\frac{n-2}{2}\right)!} a$, if n is even

$$\begin{aligned}
 I_n &= \int_0^\infty \underbrace{x^{n-1}}_I \underbrace{x e^{-x^2}}_{II} dx \\
 &= x^{n-1} \int_0^\infty x e^{-x^2} dx - \int_0^\infty (n-1)x^{n-2} \int_0^\infty x e^{-x^2} dx dx \\
 &= \left[-x^{n-1} \frac{e^{-x^2}}{2} \right]_0^\infty + \frac{(n-1)}{2} \int_0^\infty x^{n-2} e^{-x^2} dx \\
 I_n &= 0-0 + \frac{n-1}{2} I_{n-2}
 \end{aligned}$$

$\frac{x^{n-1}}{e^{x^2}}$
 $x \rightarrow \infty$

$$I_n = \frac{n-1}{2} I_{n-2}$$

$$I_{n-2} = \frac{n-3}{2} I_{n-4}$$

$$I_{n-4} = \frac{n-5}{2} I_{n-6}$$

$$I_n = \frac{n-1}{2} \cdot \frac{n-3}{2} \cdot \frac{n-5}{2} \cdots \frac{3}{2} \cdot \frac{1}{2} I_0 \quad n = \text{even}$$

$$I_n = \frac{1 \cdot 3 \cdot 5 \cdot 7 \cdots (n-5)(n-3)(n-1)}{2^{n/2}} \cdot a \frac{2 \times 4 \times 6 \cdots (n-2)}{2 \times 4 \times 6 \cdots (n-2)}$$

$$I_n = \frac{(n-1)}{\left(\frac{n-2}{2}\right) \cdot 2^{n-1}} a = \frac{n}{\left(\frac{n}{2}\right) 2^n} a$$

$2^{n/2-1}$ times

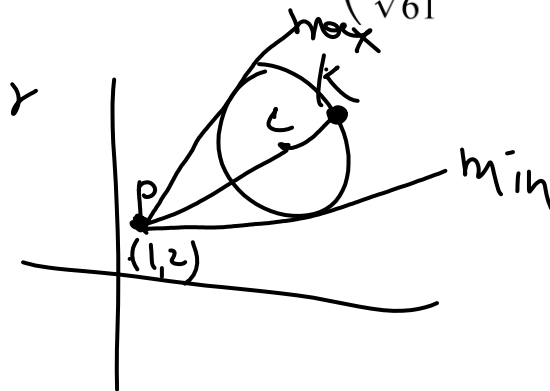
4. Pushpa is standing at (1,2) (height of pushpa is negligible). Another person Konda reddy is moving on a circular path $(x - 6)^2 + (y - 8)^2 = 4$. Pushpa wants to kill the Konda reddy by firing the bullet than which of the following is/are correct :

(Assuming path of bullet is straight line).

(Possibility of killing is inversely proportional to distance between pushpa & Konda)

- (A) The minimum angle pushpa can fire with +ve x-axis is $\tan^{-1} \left(\frac{15 + \sqrt{228}}{21} \right)$.
- ✓ (B) The maximum angle pushpa can fire with +ve x-axis is $\tan^{-1} \left(\frac{30 + \sqrt{228}}{21} \right)$.
- ✓ (C) The minimum distance between pushpa and Konda is $(\sqrt{61} - 2)$.
- ✓ (D) The least possibility to kill the Konda is when he is at the point $\left(\frac{10}{\sqrt{61}} + 6, \frac{12}{\sqrt{61}} + 8 \right)$.

min distance = cp - r
 $\sqrt{25+36} - 2$



slope of CP

$$m = \frac{6}{5}$$

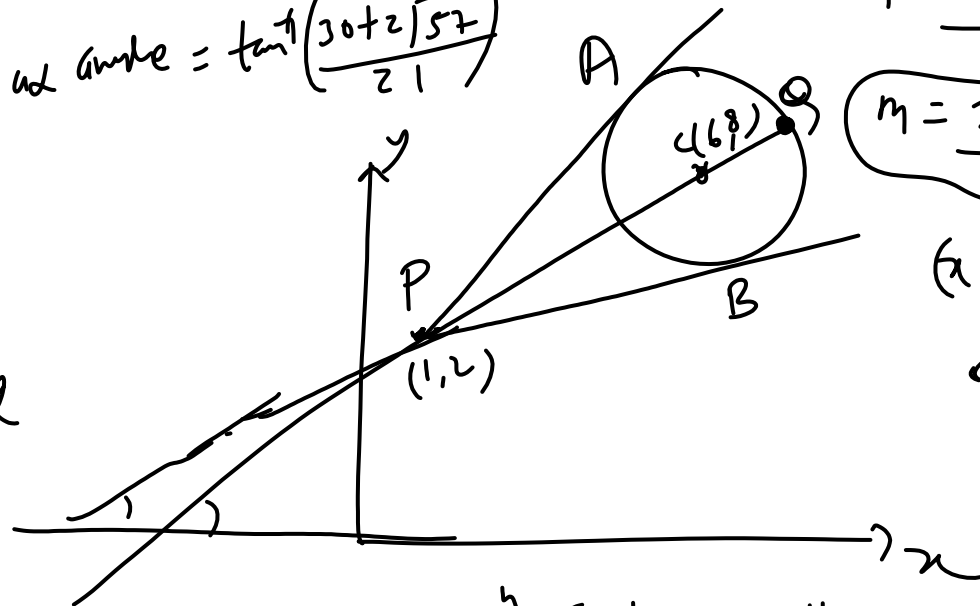
eq in parametric form
CP

$$\frac{x-6}{5/\sqrt{61}} = \frac{y-8}{6/\sqrt{61}} = 2$$

$$Q(x, y) \Rightarrow \left(\frac{10}{\sqrt{61}} + 6, 8 + \frac{12}{\sqrt{61}} \right)$$

$$\text{min angle} = \tan^{-1} \left(\frac{30 - 2\sqrt{57}}{21} \right)$$

$$\text{max angle} = \tan^{-1} \left(\frac{30 + 2\sqrt{57}}{21} \right)$$



eq of tangent through P

$$y-2 = m(x-1)$$

$$\Rightarrow mx - y - m + 2 = 0$$

$$\text{length of } \perp = 2$$

$$\left| \frac{6m - 8 - m + 2}{\sqrt{m^2 + 1}} \right| = 2$$

$$\Rightarrow 25m^2 + 3(-60m) = 4 + 4m^2$$

$$\Rightarrow 21m^2 - 60m + 39 = 0$$

$$m = \frac{60 \pm \sqrt{3600 - 30 \times 2 \times 4}}{2 \times 21}$$

$$m = \frac{60 \pm 4\sqrt{225-168}}{42}$$

$$m = \frac{30 \pm 2\sqrt{57}}{21}$$

$$(x-6)^2 + (y-8)^2 = 2^2$$

$$C(6, 8)$$

5. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$,
 $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2013)$ is greater than or equal to -

(A) 3

(C) 5

(B) 4

(D) 6

$$g(2013) = f(2013) + 5 - 2013$$

$$g(2013) = 5$$

$x \rightarrow x-4$
 $f(x+1) \geq f(x-4) + 5$

$$f(x-4) + 5 \leq f(x+1) \leq f(x) + 1$$

$$f(x-3) + 4 \leq f(x-4) + 5$$

$$f(x-2) + 3 \leq f(x-3) + 4$$

$$f(x-1) + 2 \leq f(x-2) + 3$$

$$f(x) + 1 \leq f(x-1) + 2$$

$$f(x) + 1 \leq f(x+1) \leq f(x) + 1$$

$$\Rightarrow f(x+1) = f(x) + 1$$

$$f(2) = f(1) + 1 = 2$$

$$f(3) = f(2) + 1 = 3$$

$$\vdots$$

$$f(2013) = 2013$$

6. Let a plane perpendicular to the plane $2x + 3y + 4z - 5 = 0$ contains the line $\frac{x-0}{5} = \frac{y-1}{3} = \frac{z-2}{4}$ and line L where line L is contained in the plane $2x + 3y + 4z - 5 = 0$, then :

- (A) Line L passes through point $\left(0, \frac{7}{25}, \frac{26}{25}\right)$ $\Rightarrow \frac{21}{25} + \frac{104}{25} - 5 = 0$ ✓
 (B) Line L passes through point $(0, 1, -1)$ $\Rightarrow \frac{28}{25} - \frac{178}{25} + 2 = 0$ ✓
 (C) One of the direction ratios of line L is $(-25, 6, 8)$ ✓
 (D) One of the direction ratios of line L is $(-25, -6, -8)$

D.R.s of Normal of P_2

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 3 & 4 \\ 2 & 3 & 4 \end{vmatrix} = 0\hat{i} - 12\hat{j} + 9\hat{k}$$

$\text{or } \vec{n} = 4\hat{j} - 3\hat{k}$

Eqⁿ of P_2 : $0(x-0) + 4(y-1) - 3(z-2) = 0 \Rightarrow$
 $\Rightarrow \boxed{4y - 3z + 2 = 0}$

$$\text{D.R.'s of line } L = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 0 & 4 & -3 \end{vmatrix} = -25\hat{i} + 6\hat{j} + 8\hat{k}$$

7. If foci of an ellipse be $(-1, 2)$ and $(-2, 3)$ and its tangent at a point A is $2x + 3y + 9 = 0$

✓ (A) Length of the minor axis of the ellipse will be $2\sqrt{14}$

✓ (B) Co-ordinate of the point 'A' will be $\left(\frac{-32}{9}, \frac{-17}{27}\right)$

✗ (C) Distance between the foci is $2\sqrt{2}$

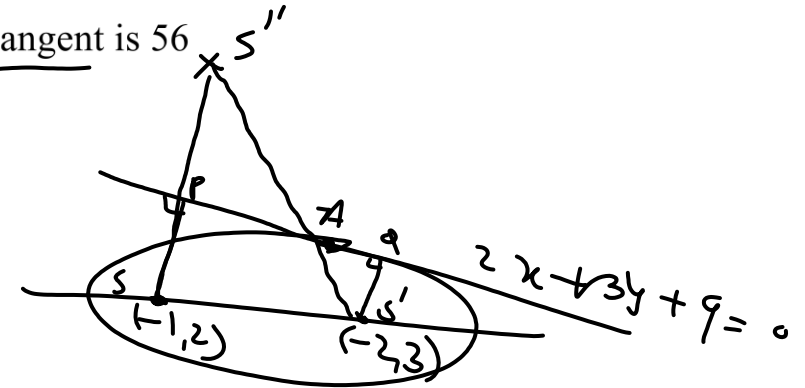
✗ (D) Product of the perpendiculars from foci to any tangent is 56

$$SP \cdot SQ = \left| \frac{-2+6+9}{\sqrt{4+9}} \right| \left| \frac{-4+9+9}{\sqrt{4+9}} \right| = b^2$$

$$b^2 = 14$$

$$b = \sqrt{14}$$

$$\text{length} = 2\sqrt{14}$$



$$SS' = \sqrt{(-2+1)^2 + (3-2)^2}$$

$$= \sqrt{2}$$

$$2x + 3y + 9 = 0$$

$$s(-1, 2)$$

$$\frac{h+1}{2} = \frac{k-2}{3} = -2 \left(\frac{-2+6+9}{-4+9} \right)$$

$$h = -5$$

$$k = -4$$

$$s''(-5, -4)$$

$$s'(-3, 3)$$

$$s' s''$$

$$y+4 = \frac{3+4}{-2+5} (x+5)$$

$$3y+12 = 7x+35$$

$$7x - 3y + 23 = 0$$

$$2x + 3y + 9 = 0$$

add

$$9x + 32 = 0$$

$$x = -\frac{32}{9}$$

8. Let α, β are non-zero roots of quadratic equation $f(x) = 0$. If $S_n = p\alpha^n + q\beta^n$ and satisfying the relation $S_{n+2} - 3S_{n+1} + 2S_n = 0 \forall n \in \{0, 1, 2, \dots\}$ then $f(x)$ can be equal to (where $p > 0, q > 0$)

~~(A)~~ $(x-1)^2$

~~(B)~~ $(x-2)^2$

~~(C)~~ $(x-1)(x-2)$

~~(D)~~ $x^2 + 3x + 2$
 $x = -1, -2$

$$S_{n+2} - 3S_{n+1} + 2S_n = 0$$

$$(p\alpha^{n+2} + q\beta^{n+2}) - 3(p\alpha^{n+1} + q\beta^{n+1}) + 2(p\alpha^n + q\beta^n) = 0$$

$$p\alpha^n (\alpha^2 - 3\alpha + 2) + q\beta^n (\beta^2 - 3\beta + 2) = 0$$

$$\alpha^2 - 3\alpha + 2 = 0$$

$$\alpha = 1, 2$$

$$\beta^2 - 3\beta + 2 = 0$$

$$\beta = 1, 2$$

$$\begin{matrix} 1, 1 \\ 2, 2 \\ 1, 2 \end{matrix}$$

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33 , $-.30$, 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The triangle ABC, right angled at 'C', has Median AD, BE and CF. AD lies along the line $y = x+3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60 and area of $\Delta ABC = 1600K$ then value of 'k' is-

$$y = x+3 \quad \left| \quad y = 2x+4 \right.$$

$$m_1 = 1 \quad \left| \quad m_2 = 2 \right.$$

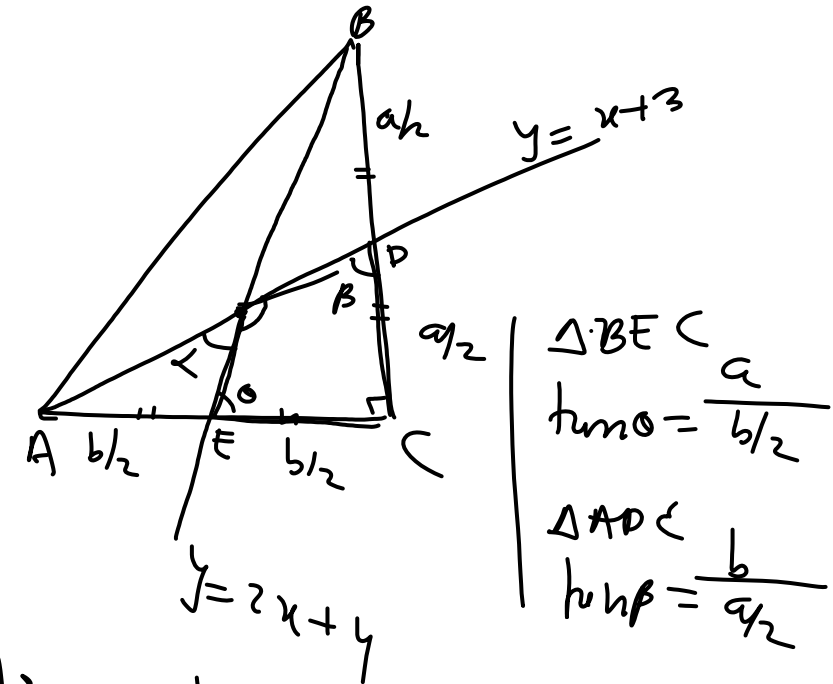
$$\tan \alpha = \left| \frac{1-2}{1+1 \times 2} \right| = \frac{1}{3}$$

$$\pi - \alpha + \beta + 90^\circ + \theta = 360^\circ$$

$$- \alpha + \beta + \theta = 90^\circ$$

$$- \alpha = 90 - (\beta + \theta)$$

$$- \tan \alpha = \cot (\beta + \theta)$$



$$\text{Area} = \frac{1}{2} \times a \times b = 1600K$$

$$\tan(\beta + 0) = -6 + x$$

$$\frac{\tan \beta + \tan 0}{1 - \tan \beta \tan 0} = -3$$

$$\frac{\frac{2b}{a} + \frac{2a}{b}}{1 - \frac{2b}{a} \times \frac{2a}{b}} = 3$$

$$\frac{2 \left(\frac{a^2 + b^2}{ab} \right)}{-3} = -3$$

$$2(a^2 + b^2) = 9ab$$

$$2 \times 60^2 = 9ab$$

$$ab = 800$$

$$\frac{1}{2} ab = 1600K$$

$$400 = 1600K$$

$$K = \frac{1}{4}$$

2. Let $2(f(x))^2 - \frac{d^2 f(x)}{dx^2} f(x) + \left(\frac{df(x)}{dx}\right)^2 = 0$ & $f(0) = f(1) = -1$. Area of region bounded by $y = 0$, $x = 0$, $x = 1$ and $y = (2x - 1)f(x)$ is $2 \left(\frac{e^{1/a} - 1}{e^{1/a}} \right)$, then 'a' is ?

$$2(f(x))^2 - f''(x)f(x) + (f'(x))^2 = 0$$

$$2 = \frac{f(x)f''(x) - (f'(x))^2}{(f(x))^2}$$

$$2 = \frac{d}{dx} \left(\frac{f'(x)}{f(x)} \right)$$

$$2x + 1 = \frac{f'(x)}{f(x)}$$

$$\int (2x + 1) dx = \int \frac{f'(x)}{f(x)} dx$$

$$x^2 + x + C = \log |f(x)|$$

$$x = 0 \quad 0 + 0 + C = \log |f(0)|$$

$$C = 0$$

$$x = 1 \quad 1 + 1 + C = \log |f(1)|$$

$$1 + 1 + 0 = 0$$

$$2 = -1$$

$$\ln |f(x)| = x^2 - x$$

$$|f(x)| = e^{x^2 - x}$$

$$f(x) = e^{x^2 - x}$$

$$= 2 \left(\frac{e^{1/4} - 1}{e^{1/4}} \right)$$

$$\underline{\underline{a = 4}}$$

$$\int_0^1 |2x-1| e^{x^2-x} dx$$

$$= - \int_0^{1/2} (2x-1) e^{x^2-x} dx + \int_{1/2}^1 (2x-1) e^{x^2-x} dx$$

$$= - \left(e^{x^2-x} \right)_0^{1/2} + \left(e^{x^2-x} \right)_{1/2}^1$$

3. Number of solution(s) of the equation $8000 \sin^3 x + 3375 \cos^3 x + 1728 = 5400 \sin 2x$ in $(0, 2\pi)$ is(are)

$$(20 \sin x)^3 + (15 \cos x)^3 + 12^3 = 5400 \sin x \cos x$$

$$a^3 + b^3 + c^3 = 3abc \Leftrightarrow \begin{matrix} a=b=c \\ \text{or} \\ a+b+c=0 \end{matrix}$$

(C.1)

$$20 \sin x = 15 \cos x = 12$$

$$\rightarrow \sin x = \frac{12}{20} = \frac{3}{5}$$

$$\rightarrow \cos x = \frac{12}{15} = \frac{4}{5}$$

$$= 3 \times 20 \sin x \times 15 \cos x \times 12$$

$$\left. \begin{array}{c} \frac{1}{\checkmark} \\ \frac{1}{\checkmark} \end{array} \right\} \text{2 soln}$$

$$c) \quad 20 \sin x + 15 \cos x + 12 = 0$$

$$\Rightarrow 5(4 \sin x + 3 \cos x) = -12$$

$$\Rightarrow 4 \sin x + 3 \cos x = -\frac{12}{5}$$

$$\Rightarrow \sin(x + \alpha) = -\frac{12}{25}$$

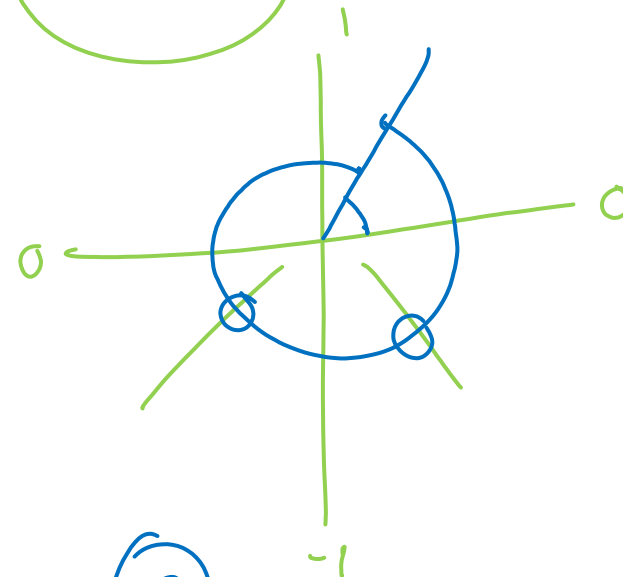
$$\therefore x \in (0, 2\pi)$$

$$x + \alpha \in (2, 2\pi + \alpha)$$

$$\sin t = -\frac{12}{25}$$

$$a \sin x + b \cos x = c$$

$$(\alpha = \frac{4}{5})$$



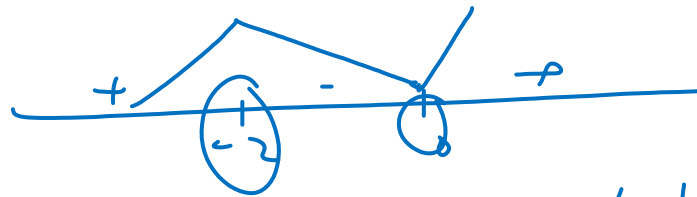
$$\rightarrow 3\pi/4$$

4. If $f(x) = x^3 + 3x^2 - 1$, then number of real solutions of $f(f(x)) = 0$, is

$$f'(x) = 3x^2 + 6x$$

$$f'(x) = 3x(x+2) = 0$$

$$\therefore x = 0, -2$$



$$x = -2$$

$$x = 0$$

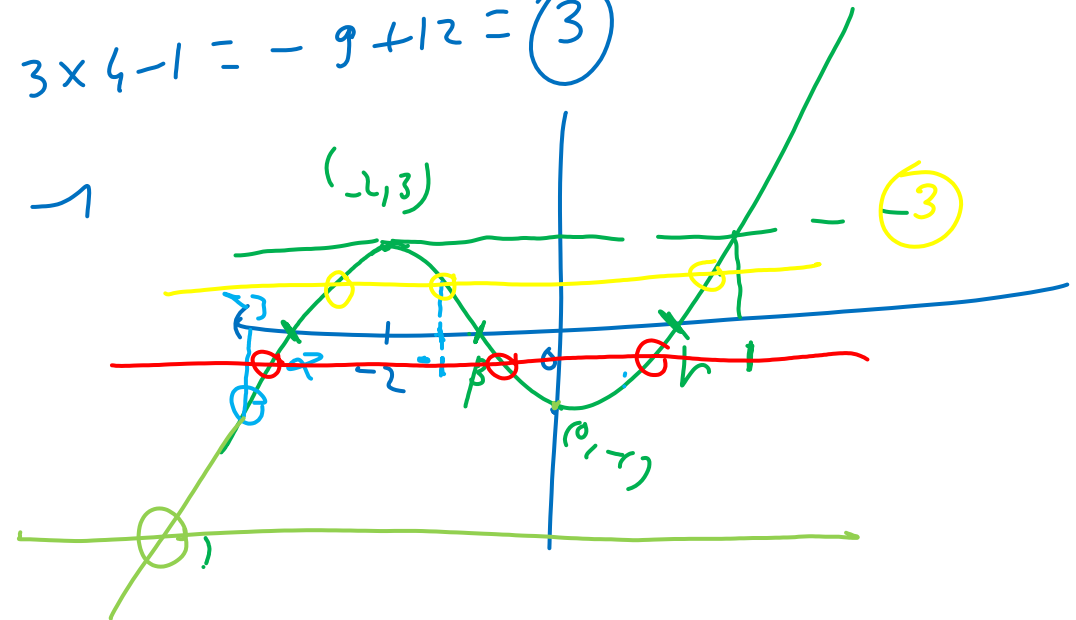
$$f(-2) = -8 + 3 \times 4 - 1 = -9 + 12 = 3$$

$$f(0) = -1$$

c.f

$$(f(x))^3 + 3(f(x))^2 - 1 = 0$$

$$\text{degree} = 9$$



$$f(x) = x^3 + 3x^2 - 1$$

$$x = -3$$

$$f = -27 + 3 \times 9 - 1 = -27 + 27 - 1 = -1$$

$$-3 < \alpha < -2 \quad f(\alpha) = 0$$

$$f(x) = \alpha \quad \text{--- 1 sol}^n$$

$y = f(x) \quad y = \alpha$

$$x = -1$$

$$f = -1 + 3 - 1 = 3 - 2 = 1$$

$$-1 < \beta < 0 \quad f(\beta) = 0$$

$$f(x) = \beta \quad \text{--- 3 sol}^n$$

$$x = 1$$

$$f = 1 + 3 - 1 = 3$$

$$0 < \gamma < 1 \quad f(\gamma) = 0$$

$$f(x) = \gamma \quad \text{--- (3)}$$

$$f(f(x)) = 0$$

$\alpha \quad \beta \quad \gamma$

$$7 \text{ sol}^n$$

5. In a $\triangle ABC$, the vertex A is (1, 1) and its orthocenter is (2, 4). If the sides AB and BC are members of the family of straight lines $ax + by + c = 0$, where a, b, c are in A.P., then the coordinates of vertex C are (h, k). The value of $2h + 12k$ is equal to _____.

$$2 \times -17 + 12 \times 4$$

$$= -34 + 48$$

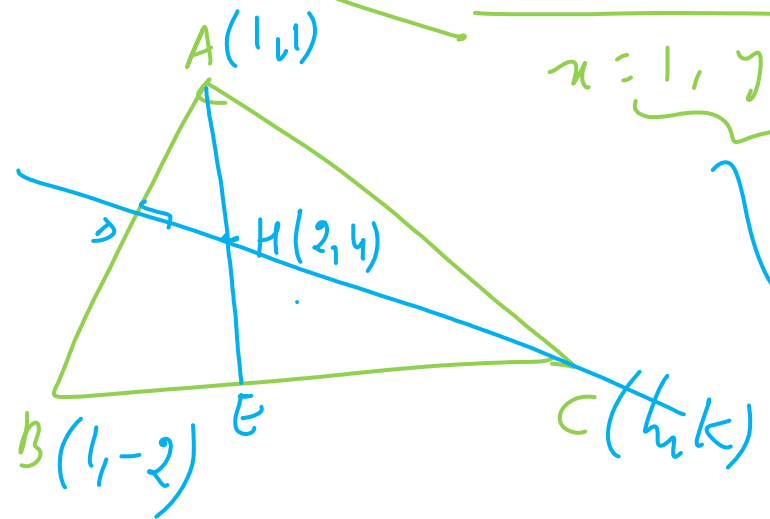
$$= 14$$

AB and BC

$$\begin{cases} ax + by + c = 0 & \text{--- (i)} \\ a - 2b + c = 0 & \text{--- (ii)} \end{cases}$$

$$2b = a + c$$

$$x = 1, y = -2$$



$$m_{AB} = \frac{3}{6} = \text{N.D.}$$

$$m_{BC} = 0$$

$$\therefore \text{we have } y = 4 \text{ --- (1)}$$

$$\therefore k = 4$$

$$A(1,1), H(2,4)$$

$$m_{AH} = \frac{3}{1}$$

$$m_{BC} = -\frac{1}{3} \quad B(1, -2)$$

$$BC: y + 2 = -\frac{1}{3}(x - 1)$$

$$\therefore 6 = -\frac{1}{3}(x - 1)$$

$$\Rightarrow x - 1 = -18$$

$$\boxed{x = -17 = h}$$

6. A line L in the Argand plane is called as the mean line for points $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$, if there exist points $\beta_1, \beta_2, \beta_3, \dots, \beta_n$ on line L such that $\sum_{r=1}^n (\alpha_r - \beta_r) = 0$. If the line L passes through the point $-5 + 6i$, then the slope of the mean line L for points $-2 + 3i, 5 + 8i, -6 + 60i, 5 - 581i$ is λ , then value of $[[\lambda]]$ is : (Where $[.]$ greatest integer function)

$$\sum_{r=1}^n \alpha_r = \sum_{r=1}^n \beta_r$$

$$= 2 - 510i$$

$$\begin{array}{r} -8+10 \\ \text{---} \\ 2 \end{array} \quad \begin{array}{r} 71 \\ -581 \\ \text{---} \\ -510 \end{array}$$

$$\vec{r} = \vec{a} + \lambda \vec{p}$$

$$L \rightarrow (-5+6i) \quad a+ib$$

$$Z = (-5+6i) + t(a+ib)$$

$$L \rightarrow \text{slope} = \frac{b}{a}$$

$$\begin{array}{r} 22 \overline{) 534} \overline{) 24.1} \\ \underline{44} \\ 94 \\ \underline{88} \\ 6 \end{array}$$

$$\sum_{r=1}^4 p_r = -5 \times 4 + 4 \times 6i + K(a+ib)$$

$$\begin{aligned} 2 - 510i &= -20 + 24i + K(a+ib) \\ &= (-20 + Ka) + (24 + 6K)i \end{aligned}$$

$$\begin{aligned} \Rightarrow -20 + Ka &= 2 \\ Ka &= 22 \text{ --- (I)} \end{aligned}$$

$$\begin{aligned} 24 + 6K &= -510 \\ 6K &= -534 \text{ --- (II)} \end{aligned}$$

$$\frac{b}{a} = -\frac{534}{22} = -24.1$$

$$[\lambda] = -25 \quad \boxed{4|[\lambda]| = 25}$$

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